

FOCAL ARTICLE

75 Years After Likert: Thurstone Was Right!

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Abstract

For over three-quarters of a century researchers and practitioners have analyzed rating scale data using methods that assume a dominance response process wherein an individual high on the trait assessed is assumed to answer positively with high probability. This approach derives from Likert's famous 1932 approach to the development and analysis of rating scales. In this paper, we argue that Likert scaling and related methods are misguided. Instead, we propose that methods that have evolved from Thurstone (1927, 1928, 1929) scaling provide a better representation of the choice process underlying rating scale judgments. These methods hypothesize an ideal point response process where the probability of endorsement is assumed to be directly related to the proximity of the statement to the individual's standing on the assessed trait. We review some research showing the superiority of ideal point methods for personality assessment and then describe several settings in which ideal point methods should provide tangible improvements over traditional approaches to assessment.

In a series of remarkable papers in the late 1920s, Louis Thurstone asserted "Attitudes can be measured" (1928); see also Thurstone (1927, 1929). Central to his approach was the assumption that a conscientious person would endorse a statement that reflected his or her attitude, but "as a result of imperfections, obscurities, or irrelevancies in the statement, and inaccuracy or carelessness of the subjects" not everyone

would respond accurately (1929, p. 224). Using Thurstone's notation, suppose there were N_1 people with an attitude value of S_1 ; all should endorse a statement with scale value S_1 if they were conscientious and the item was perfect. In practice, Thurstone expected only n_1 to agree with the statement, where $n_1 < N_1$. Moreover, these people would endorse another statement with scale value S_2 (where $S_2 \neq S_1$) with probability p that is inversely related to $|S_2 - S_1|$. Figure 1, from Thurstone's 1929 paper, illustrates his theory.

In his 1928 paper, Thurstone used the example of a militarism–pacifism attitude with six statements representing a range of attitudes. Figure 2, from Thurstone (1928, p. 537), gives the locations of the six statements and shows the distribution of

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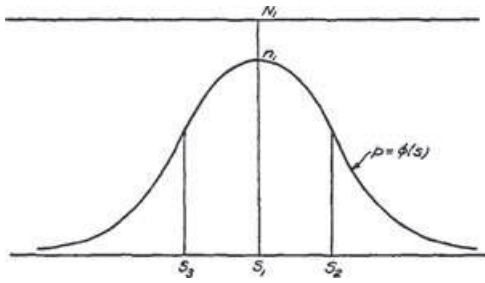


Figure 1. Response probabilities from Thurstone (1929). Reprinted from Thurstone (1929, p. 229).

the attitude in some population. Thurstone noted a pacifist "would be willing to indorse all or most of the opinions in the range d to e and ... he would reject as too extremely pacifistic most of the opinions to the left of d , and would also reject the whole range of militaristic opinions" (p. 539). Of critical import is Thurstone's method of scoring: A person's "attitude would then be indicated by the average or mean of the range [of statements] that he indorses" (p. 539). For example, Person 1 might endorse statements f and d from Figure 2, Person 2 might endorse e and b , and Person 3 might endorse c and a . Although each endorses two items, their attitudes are quite different. By attending to *which* items are endorsed and not simply how many, Thurstone's scoring allows individuals who endorse the same number of items, but who have different attitudes, to be differentiated.

In 1932, Likert provided a much simpler alternative to Thurstone scaling. Although he examined several approaches, Likert found that using a 5-point response

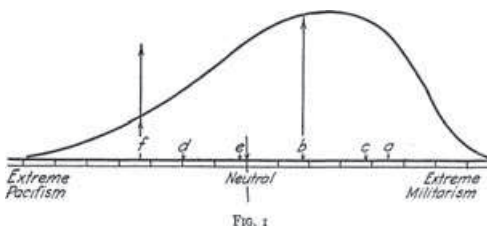


Figure 2. Locations of six militarism-pacifism attitude statements. Reprinted from Thurstone (1928, p. 537).

scale with options "Strongly Approve," "Approve," "Undecided," "Disapprove," and "Strongly Disapprove," and integer scoring ("Strongly Approve" = 5, ..., "Strongly Disapprove" = 1), yielded "the same reliability with fewer items, or higher reliability with the same number of items" as Thurstone scaling (p. 34). Using an Internationalism scale as an example, Likert assigned the largest value of the response scale "to the end [of each statement] which seemed to favor internationalism" and thereby invented reverse scoring of negatively phrased items. After reverse scoring, an individual's total score could be taken as the sum or mean of the item scores.

A crucial difference between Likert and Thurstone concerns intermediate statements like "Compulsory military training in all countries should be reduced but not eliminated" (p. 34). Likert argued that "It is impossible to tell whether a person is agreeing or disagreeing with the 'reduction' aspect of this statement or the 'not eliminated' aspect" and therefore this "statement is double-barreled and of little value because it does not differentiate persons in terms of their attitudes" (p. 34). Consequently, Likert recommended deleting items like this intermediate statement. Thurstone, on the other hand, viewed this statement, like statement e in Figure 2, as necessary for accurately measuring the attitudes of people with intermediate standings. Therefore, Thurstone deliberately wrote intermediate items and included them in his measures.

Likert based his conclusion that double-barreled items were worthless on item-total correlations. He argued that "If a zero or very low correlation coefficient is obtained, it indicates that the statement fails to measure that which the rest of the statements measure" (p. 48) and "Thus item analysis reveals the satisfactoriness of any statement so far as its inclusion in a given attitude scale is concerned" (p. 49).

Although Likert did not articulate a psychometric model for his procedure, his approach implies what Coombs (1964) called a *dominance response process*. Here an individual high on the trait or

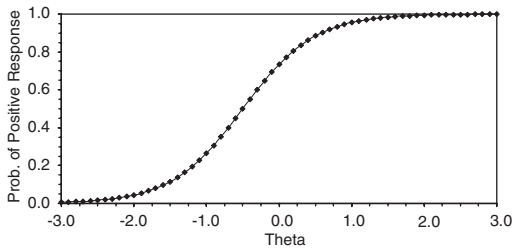


Figure 3. Probability of a positive response to a dichotomously scored statement as a function of the latent attitude (theta).

attitude measured by a scale is likely to “Strongly Agree” with a positively worded statement and “Strongly Disagree” with a negatively worded statement. Figure 3 illustrates a dominance response process for a dichotomously scored item. Note that as the attitude—labeled theta in Figure 3—increases, so does the probability of a positive response.

We believe that dominance models are most sensibly applied to domains in which an individual’s capacity or maximum performance capability is pitted against the difficulty or extremity of an item (Stark, Chernyshenko, Drasgow, & Williams, 2006; Tay, Drasgow, Rounds, & Williams, 2009). Consider a weight lifter attempting to clean and jerk a series of increasing weights. The weight lifter’s strength would be denoted as theta in Figure 3 and a response function—the smooth curve in the figure—could be created for each weight. A stronger weight lifter would be indicated by a theta value further to the right on the figure and he or she would have a higher probability of successfully lifting the weight.

Much of the work on dominance models has been in the context of cognitive ability testing. Here, an individual’s ability is assessed by a set of items of varying difficulty. An individual with a high ability level is expected to answer all the easy items correctly, all the moderately difficult items correctly, and some of the most difficult items correctly. Thus, the individual *dominates* the easy and moderately difficult items.

We argue that psychometric models for dominance response processes, such as classical test theory, factor analysis, and the logistic item response theory (IRT) models, are ill suited for response processes requiring introspection (Chernyshenko, Stark, Drasgow, & Roberts, 2007; Tay et al., 2009). When individuals consider their militarism–pacifism or internationalism, we believe that they ask “Does this statement closely describe me?” The closer an item’s location on the attitude continuum to the individual’s location on that continuum, the greater the probability that the person will endorse the item. The maximum probability of endorsement occurs when the attitude level of the item equals the individual’s attitude level. This is the idea underlying Thurstone’s model shown in Figure 1. Coombs (1964) is credited with coining the term “ideal point” and wrote, “We conceive, then, of representing an individual by a point in the same space containing the stimulus points, in such a way that the point corresponding to the individual is a point of his maximum preference in this domain of stimuli” (p. 8). Coombs used the term “unfolding technique” for formal representations of this process because the probability of endorsement decreases in both directions from the individual’s ideal point: statements representing lower and higher locations on the latent trait continuum have decreasing probabilities of being endorsed as they are further away from the individual’s ideal point.

Figure 4 presents a more modern formulation of the Thurstone’s model shown in Figure 1. Here, the probability of a positive response to a dichotomously scored item is given as a function of the trait or attitude it assesses; this curve is called an item response function. For example, an extraversion scale might contain the item “I enjoy chatting quietly with a friend at a café.” As shown in Figure 4, individuals who are too introverted might tend to disagree with the item because they are uncomfortable in public places. In direct contrast to a dominance response process, individuals who are high on the trait

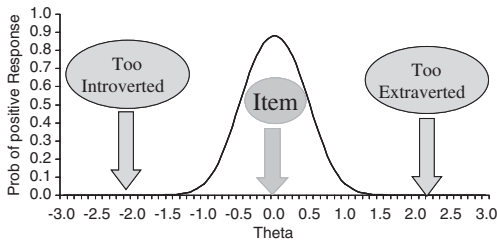


Figure 4. Item response function for an ideal point response process.

assessed—extraversion in Figure 4—also tend to disagree because they prefer more exciting settings.

What is the item–total correlation of items of the sort depicted in Figure 4? Individuals who are low in extraversion would tend to receive scores of 0 on this dichotomously scored item and individuals who are high in extraversion would also tend to receive 0's. Only individuals who are intermediate would tend to receive 1's. There is little *linear* trend in the item score (but certainly a nonlinear trend) as a function of the trait assessed and therefore the item–total correlation would be close to zero. Is this a bad item? No! It provides useful information about the trait assessed—a score of 1 indicates an intermediate degree of extraversion, whereas a 0 indicates high or low (but not intermediate) extraversion. We believe Likert was misled because, as is so common in psychology, he looked only for linear relations.

Personality Assessment and Ideal Point Models

In the late 1990s, we began fitting IRT models to data from personality scales. Based on earlier work by Reise and Waller (1990), we expected the two-parameter logistic model (2PLM) to fit well. To our surprise, fits of the 2PLM and other IRT models were noticeably worse than the fits of IRT models to data from cognitive ability tests. In retrospect, we should not have been surprised because personality items are essentially attitude statements about oneself (Chernyshenko, Stark, Chan, Drasgow, & Williams, 2001)

and Thurstone's work on attitude measurement should be directly applicable to personality.

In our research, we have used a chi-square goodness-of-fit measure that compares the observed and expected (on the basis of the IRT model) frequencies of endorsing/not endorsing items. This measure can be computed for individual items, pairs of items, and triples of items. For example, the chi-square for a pair of items compares the observed and expected frequencies in the two-way table formed by crossing endorse/not endorse on the first item with endorse/not endorse on the second item (see Drasgow, Levine, Tsien, Williams, & Mead, 1995 for details).

Interestingly, we found (Chernyshenko et al., 2001) that IRT models—even misspecified ones—do a good job of reproducing the observed frequencies of single items. The chi-squares for pairs and triples of items present a more challenging test of the fit of the IRT model (we have found that higher order tables have less power to detect misfit, presumably because the sample sizes in some cells become too small).

When an IRT model is estimated using data from a large sample (i.e., 3,000 or more) and fit is evaluated in a cross-validation sample, we've found that a chi-square to degrees of freedom (*df*) ratio of less than 2 indicates an excellent fit, between 2 and 3 indicates a generally satisfactory fit, and over 3 indicates misfit.

Sixteen Personality Factor (16PF) Questionnaire

Chernyshenko et al. (2001) fit several IRT models to data from the 16PF (Conn & Rieke, 1994). Although they analyzed all 16 scales, we shall discuss just the Sensitivity scale. Data from 6,455 individuals were used to estimate item parameters and 6,456 individuals served as a cross-validation sample. The mean chi-square to *df* ratio for single items was 0.98 for the 2PLM, but was 4.05 and 5.45 for pairs and triples of items, which is clearly unsatisfactory. The three-parameter logistic model (3PLM) did little

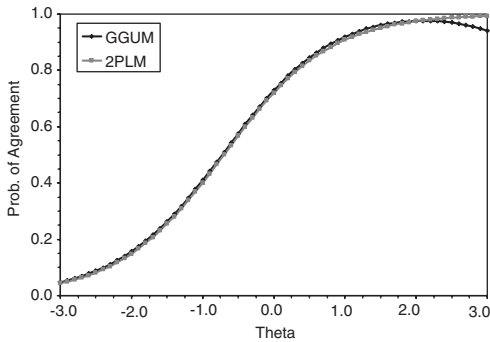


Figure 5. Two-parameter logistic model and generalized graded unfolding model item response functions for the same item.

better, with 0.87, 3.89, and 5.23 for singles, doubles, and triples of items, respectively. These findings led us to question the appropriateness of the logistic IRT models for personality data.

We also fit Levine's (1984) maximum likelihood formula score model (MFSM). MFSM is a nonparametric IRT model and thereby provides great flexibility: The best fitting item response function need not be logistic or even monotonic. Although the fit of this model to individual items was barely satisfactory (mean chi-square to *df* ratio of 2.91), the fit to doubles and triples of items was noticeably better (2.61 and 2.42, respectively) than the two logistic models. Interestingly, some of the MFSM item response functions showed nonmonotonicity: The probabilities of endorsement did not always increase, which is the hallmark of dominance models. Instead, the probability of endorsement increased in one part of the trait continuum but *decreased* in another part. We are indebted to Michael Levine for pointing out that this nonmonotonicity is suggestive of an ideal point response process.

Developers of personality scales compute item-total correlations, internal consistency reliability, and factor loadings and delete items with poor psychometric properties according to these dominance model analyses. Consequently, it was surprising to find evidence of unfolding (i.e.,

endorsement probabilities that increased at lower trait levels and then decreased at higher trait levels) with the 16PF scales. Note that by deleting intermediate items like the one depicted in Figure 4 and reverse scoring negative items, we should be left with items that ought to be fit well with the 2PLM: Figure 5 shows endorsement probabilities estimated with an ideal point model (the generalized graded unfolding model [GGUM] developed by Roberts, Donoghue, & Laughlin, 2000) and the 2PLM. These item response functions were estimated for the item "Even when something bad happens, I can push negative thoughts out of my mind" from the Well-Being scale of the Tailored Adaptive Personality Assessment System (TAPAS; Drasgow, Chernyshenko, & Stark, 2010). These two item response functions are virtually identical over almost all the trait continuum, differing appreciably only at trait values above 3.0 (IRT trait scores are given as standardized scores). Because these two item response functions are virtually identical, they cannot be differentiated on the basis of a goodness-of-fit measure.

Constructing New Personality Scales

Table 1 briefly summarizes instrument development for dominance and ideal point models. Note that there are important differences at each step. For example, within the ideal point framework, scores could be computed as the mean of the item locations of the items endorsed, rather than the proportion endorsed or sum of item scores as commonly used within the dominance framework.

To better understand the characteristics of items that have *not* been preselected by dominance model methods, Chernyshenko et al. (2007) reported the analysis of responses from 539 students to 50 new single-statement personality items written for the Order facet of the conscientiousness Big Five personality dimension. Items were written to assess the entire range of this dimension: low, intermediate, and high. For dominance model analyses, items written

Table 1. *Comparison of Dominance Model and Ideal Point Model Procedures*

Activity	Dominance model	Ideal point model
Item development	Write positive and negative items	Write positive, intermediate, and negative items
Item scoring	Reverse score negative items	Do not reverse score
Item analysis	Compute item–total correlations	Compute item–subtotal correlations ^a
IRT analysis	Use a logistic model	Use an ideal point model
Trait scores	Compute the proportion endorsed, the sum of item scores, or the logistic model trait estimate	Compute the mean item location of the items endorsed or the ideal point model trait estimate

^aSubtotal scores for the negative, intermediate, and positive items can be computed and then correlations of each negative item with the negative item subtotal can be computed, correlations of the intermediate items with the intermediate item subtotal can be computed, and correlations of the positive items with the positive item subtotal can be computed.

to assess low trait values were reverse scored. Using an item–total correlation of .3 as a criterion, which was suggested by Nunnally and Bernstein (1994) as the cutoff for item retention, 13 items would have been rejected. Importantly, most of those items were designed to assess intermediate levels of orderliness. In contrast, when the items were analyzed with an ideal point IRT computer program (GGUM2000 developed by Roberts, 2001), only two items were found to have unsatisfactory discrimination parameter estimates (i.e., below 0.4). Thus, virtually all the intermediate items rejected in the dominance model analyses were found to have good psychometric properties in the ideal point analysis.

We have now developed personal-ity statements representing 22 facets of the Big Five dimensions. The statements can be administered in various formats using the TAPAS software as an on-line computerized adaptive test. Results for the Well-Being facet of the emotional stability Big Five dimension are typical. Drasgow, Chernyshenko, and Stark (2009) reported findings for 20 single-statement items completed by 445 Army recruits. The items were administered using a 4-point response scale with 1 = "Strongly Disagree," 2 = "Disagree," 3 = "Agree," and 4 = "Strongly Agree."

The five items designed to assess low well-being had a mean factor loading of

–.50 before reverse scoring and a mean corrected item–total correlation of .41 after reverse scoring. Nine items intended to assess high levels of well-being had a mean factor loading of .52 and a mean corrected item–total correlation of .46. In contrast, the six items assessing intermediate levels of well-being (e.g., "My life has had about an equal share of ups and downs") had a mean factor loading of .02 and a mean item–total correlation of .07. Clearly, these items would have been deleted if dominance model methods were used. However, their mean GGUM2000 item discrimination was 0.87 after dichotomously rescoring the items, which is a bit lower than the 1.09 mean of the negative items and the 1.35 mean of the positive items, but is nonetheless quite satisfactory.

When the fits of the models were examined, we obtained an excellent 0.71 mean adjusted chi-square to *df* ratio for the GGUM analysis of pairs of dichotomously scored items. In contrast, the mean adjusted chi-square to *df* ratio was 2.39 for pairs of items when the 2PLM was used. It is important to note that the 2PLM failed to fit in a predictable and systematic way: It was unable to accurately model the responses to the intermediate items. When a 2 × 2 table delineating positive versus negative responses for two intermediate items were constructed, the 2PLM predicted that about 25% of the respondents would fall into each

cell. However, the observed frequencies in the positive/positive and negative/negative cells were much higher than 25% because people intermediate in well-being tended to agree or strongly agree with both items and people low or high in well-being tended to disagree or strongly disagree with both.

If the six intermediate items were deleted, the fits of the 2PLM and GGUM would be very similar. That is because item response functions (IRFs) for high and low well-being items are virtually indistinguishable across the observed ranges of the trait continuum—Figure 5 presents IRFs for one such item. As can be seen, the expected endorsement probabilities for the two models are nearly the same for all but extremely positive trait levels, perhaps reflecting the fact that highly optimistic individuals do not admit experiencing “bad” events and may begin to disagree with the item. This is reflected in the gradual descent of the GGUM IRF at trait values of 2.2 and higher. However, because the number of such individuals is very small, it has little material effect on trait estimation or fit.

Despite the fact that this item has similar IRFs under the two models, the terminology used to describe the item properties differs depending on which model is used. The GGUM location parameter for this item is 2.12 indicating that it is positive in location and wording. In ideal point terminology, item location refers to the point on the trait continuum where the probability of endorsement is highest. In Thurstone’s Law of Comparative Judgment (1927), this is the scale value of the item. On the other hand, the 2PLM location parameter or, as it is often called, the “difficulty” parameter is -0.70 . The term comes from cognitive ability terminology and indicates the point on the trait continuum where the probability of a correct response is .50. Readers unfamiliar with the applications of dominance IRT models in noncognitive assessment may erroneously assume that the item is negatively worded. In fact, all we can say is that the item is “easy” (i.e., individuals with trait levels of -0.70 are expected to endorse the item with a probability of 50%). As was

noted by Chernyshenko et al. (2007), dominance model difficulty parameters (as well as p values) do not have easily understood relationships with item content in noncognitive domains.

Why Does the Choice of Psychometric Model Matter?

Currently available psychometric models for ideal point data are considerably more complicated than corresponding models for dominance data; see, for example, Equation 7 in Roberts et al. (2000, p. 6). Thus, it is reasonable to ask whether there are any tangible benefits that accrue from this added complexity. Wouldn’t it be better just to delete intermediate statements, as we have done for the past 75 years, and work with easily scalable positive and negative statements?

A first answer is that from the perspective of basic science it is important to understand how and why people respond to assessment instruments. Understanding how people answer such assessments provides us with deeper insights into the nature of their responses. In this paper and elsewhere we have argued that responses to questions requiring introspection involve a comparison process. The individual considers his or her behaviors, attitudes, feelings, or whatever is assessed and then considers what the item asks. The decision to endorse or not endorse the item is then, we believe, driven by the psychological distance between the self-perception and the perception of the statement. A formal psychometric model of this process was given by Zinnes and Griggs (1974).

A second answer is that applying the wrong measurement model can lead to mistakes. For example, Davison (1977) showed that factor analysis of a unidimensional set of ideal point items produces two factors. Questions about the latent structure of emotion, comparisons of the Big Five versus the HEXACO model (Ashton & Lee, 2007), and other debates about constructs assessed by introspection may be clouded by the applications of misspecified models.

Kurt Lewin wrote “There is nothing so practical as a good theory” (1951, p. 169) and we believe that there are numerous important implications of and applications for a good measurement theory for self-report data.

As a first application, consider the development of a new assessment tool. With an ideal point perspective, item writers can intentionally write items to assess low, intermediate, and high trait values and thus create an instrument that provides excellent measurement precision across a broad range of the latent trait continuum. In contrast, item writers do not create intermediate items when working within the dominance framework because they know that low item–total correlations will result. Paradoxically, it turns out that intermediate statements that are shunned in dominance frameworks tend to improve measurement at high and low trait values. Intuitively, one can see that adding an intermediate statement, such as “My life has had about equal amounts of ups and downs,” allows us to separate very high and very low well-being individuals from everyone else, because those individuals would disagree with the statement. This was illustrated by Chernyshenko et al. (2007) for the Order facet of conscientiousness: A substantial gain in test information, and a corresponding reduction in the conditional standard error of measurement, was obtained for a wide range of trait values. In research contexts where the focus is primarily on correlation coefficients, all that is needed is a rough separation of respondents into high and low trait groups, so traditionally used scales are more than adequate. In some important applications (e.g., personnel selection), however, correct rank ordering of individuals at the extreme trait ranges is of critical concern, so improving measurement precision would be beneficial.

Second, in our view, some constructs in organizational psychology might be better studied by embracing an ideal point perspective. For example, central to the notion of person–organization (P–O) fit is a comparison process involving what a person

desires and what an organization provides. Interest usually revolves around the gaps between P and O on various dimensions. The magnitudes and directions of these gaps can, of course, be gauged by measuring personal needs and organizational supplies via separate assessments and deriving scores through simple mechanical (e.g., profile similarity correlations) or more complex statistical methods (e.g., polynomial regression [Edwards, 1994]). But an alternative is to write statements that explicitly capture the magnitude and direction of differences between P and O along various dimensions and scale the statements and persons for each dimension separately using unidimensional ideal point models (Chernyshenko, Stark, & Williams, 2009). For example, the statement “I wish I had more autonomy at my current workplace” reflects a positive gap between P and O on the fit dimension of autonomy, whereas the statement “Managers give way too much freedom to employees here” reflects a negative gap. Furthermore, the statement “The amount of autonomy I get here is just perfect; I don’t need any more or any less” reflects near perfect congruence between personal needs and organizational provisions. Importantly, by scaling such statements using ideal point models, the resulting P–O fit scores have interpretations that are consistent with organizational theory; a score of 0 reflects excellent fit, whereas positive or negative scores indicate varying degrees of misfit.

Another domain where an ideal point perspective makes sense is performance appraisal. Statements describing low, moderate, and high levels of employee performance can be written and a rater can be asked to endorse statements that accurately describe an employee’s performance level. Here the employee’s performance would constitute the latent trait that we wish to estimate, and we believe that an ideal point response process describes the way in which a rater chooses which statements to endorse.

Borman et al.’s (2001) computerized adaptive rating scale (CARS) provides a

good example of the use of an ideal point model for performance ratings. Raters are given two statements reflecting different levels of performance and asked to choose the one that is a better description of the ratee. Zinnes and Griggs' (1974) ideal point model is used to describe the judgment process: Raters are assumed to compare each statement to the ratee's performance and then select the statement that is perceived to be closer. Borman et al. found substantially lower standard errors of measurement and higher validity for CARS in comparison with performance ratings obtained from graphic rating scales and behaviorally anchored rating scales.

The fourth, and perhaps most exciting application of ideal point models, is their use in conjunction with personality assessment instruments using forced-choice response formats.¹ Interest in forced-choice formats has been rejuvenated by relatively recent research suggesting that they have criterion-related validity and may be less susceptible to rater biases and response distortions commonly associated with Likert-type scales (Jackson, Wroblewski, & Ashton, 2000; see also Christiansen, Burns, & Montgomery, 2005; McCloy, Heggstad, & Reeve, 2005; Vasilopoulos, Cucina, Dyomina, Morewitz, & Reilly, 2006). For example, traditional single-statement items are often transparent and, when there are strong incentives to fake, respondents have been shown to increase their scores by as much as 1.5 standard deviations (Hough, Eaton, Dunnette, Kamp, & McCloy, 1990; Stark, Chernyshenko, Chan, Lee, & Drasgow, 2001). Forced-choice measures, on the other hand, typically consist of two or more statements that are matched in terms of social desirability and often the items are multidimensional. For example, respondents might be asked to "choose the

statement that is more like you" given a pair like:

- I get along well with others.
- I always get my work done on time.

By using this approach, not only is it harder to discern the correct answers (they may depend on the application), but also it is more difficult to raise substantially one's scores on all dimensions simultaneously.

Forced-choice measures involving items composed of pairs or tetrads of statements have been used for noncognitive assessment in the past (Edwards, 1954; Rounds, Henly, Dawis, Lofquist, & Weiss, 1981; White & Young, 1998), but difficulties arise. For example, the use of paired comparison items with traditional scoring is problematic because the results are ipsative (i.e., the total score on the assessment is the same for each respondent), which causes a variety of difficulties (Meade, 2004). Consequently, researchers had to rely on heuristics in scale construction or data coding to introduce variability into scale scores for interindividual comparisons in selection applications (Chernyshenko et al., 2009; Meade, 2004). To provide a model-based solution to these problems, we have developed and applied IRT models (Stark, Chernyshenko, & Drasgow, 2005; Stark & Drasgow, 2002) for paired comparison judgments.

Our IRT model for the multidimensional pairwise preference (MDPP) format is called the multi-unidimensional pairwise preference model (MUPPM; Stark, 2002; Stark et al., 2005). It asserts that people first decide whether each of the two statements describes them. This first step is modeled by the GGUM (Roberts et al., 2000). Then, if one statement or the other (but not both) is judged to describe them, they select that statement. If both or neither statement is chosen, respondents reconsider each statement in a way akin to Thurstone's (1927) discriminial process, until exactly one statement is perceived as describing them. Our simulation and empirical studies (Chernyshenko et al., 2009; Stark et al.,

1. Of course, it would be possible to use a dominance model to describe the response process underlying forced-choice judgments. However, model misspecification in this context might lead to serious problems.

2005) have shown that accurate latent trait estimates (i.e., normative scores) can be obtained when analyzing pairwise preference responses with MUPPM.

The use of IRT models capable of adequately representing forced-choice statement selection opens up a gamut of intriguing possibilities for developing new kinds of assessment instruments. For example, as the number of statements in a testing pool expands, the number of forced-choice items that can be formed increases exponentially, which is ideal for computerized adaptive testing (CAT). Thus, adaptive testing with the MDPP format enables modest-sized pools of statements (e.g., 40 or 50 per personality facet) to generate tens of thousands of items, even with matching constraints on statements' social desirability and location parameters. As with traditional IRT models, simulation studies have shown that CAT greatly reduces the number of MDPP items needed to achieve good measurement precision relative to nonadaptive tests (Stark & Chernyshenko, 2007). Importantly, having large numbers of potential pairings makes test compromise less of a concern in unproctored web-based testing environments.

Currently, in cooperation with the Army Research Institute, the Military Enlistment Processing Command, and the Defense Manpower Data Center, a personality test battery composed of MDPP items is being evaluated for military enlistment screening. Applicants for enlistment in the U.S. Army and U.S. Air Force as well as active duty personnel have taken either a paper-and-pencil version or a computer-adaptive version of the test and are being tracked on a variety of criterion measures to see if personality dimensions can add incremental validity to the existing selection process. Because parts of the sample have taken the test under operational conditions, it will be possible to evaluate the feasibility of using MUPPM in high-stakes testing environments.

Final Thoughts

Seventy-five years after Likert wrote his 1932 paper we believe there is compelling

evidence that his approach does not do justice to the underlying processes by which people make introspective judgments. Certainly, as a rough and ready approach, a Likert scale works well. But for research and applications requiring a high fidelity representation of choice processes, the Likert approach has shortcomings.

As psychologists develop assessment tools needed for their research, it seems important for the approach to measurement to be in harmony with respondents' decision processes. P-O fit or performance appraisals provide good examples of domains where the approach to measurement can be designed to be consistent with the way people make judgments. This should make responding to questionnaires easier and more straightforward for research participants and thereby improve the quality of data that are collected.

In addition to making the task for respondents easier, researchers should be able to design better assessment tools. It has been 30 years since Frederic Lord (1980) published his seminal *Applications of Item Response Theory to Practical Testing Problems* and, today, it would be difficult to find a cognitive ability test that did not use a 3PLM or a Rasch model for its design, administration, or scoring. Perhaps, in another 30 years, the same will be said of the use of ideal-point IRT models in testing domains requiring introspection (e.g., personality, values, or performance).

In addition to the personality, P-O fit, and job performance variables described earlier, ideal point models may be usefully applied to many other important variables in industrial and organizational psychology. For example, job satisfaction, organizational commitment, leader behavior, subjective well-being, perceived organizational support, and many other variables may be fruitfully conceptualized and assessed via ideal point models.

In sum, we believe that using the right measurement model holds great promise for improved research and practice. Ideal point modeling should foster improved instruments, more straightforward linkage

between item content and psychometric parameters, and sophisticated applications such as CAT. Of course, many issues and problems remain to be solved; psychometricians have devoted great effort to dominance models during the past 100 years. In contrast, just a few psychometricians have worked on ideal point models and there have been few applications to applied measurement problems. The opportunity for new and creative research is enormous.

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Developing and Scaling Personality Measures: Thurstone Was Right – But So Far, Likert Was Not Wrong

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Although we are in agreement with Drasgow, Chernyshenko, and Stark (2010) on the appropriateness of considering ideal point models in psychological testing, we focus on a number of questions that need to be addressed before concluding that the theoretical appeal of the ideal point model translates into a consistent empirical advantage to organizations that use personality tests.

Ideal Point Models Add Complexity

To date, the dominance model has dominated the thinking of test developers. In particular, both the dominance model and Rensis Likert together have created a prescription for constructing personality test items worded at the extremes of the trait continuum. For instance, the items “I am always neat and tidy” and “I am considered a very messy person” are two items reflecting high and low levels of order, respectively. When items are

worded at the extremes like this, both the dominance response model and the ideal point model yield essentially the same personality trait scores, as Drasgow et al. emphasize (also see Chernyshenko, Stark, Drasgow, & Roberts [2007] who found $r = .92$ for order; similarly, Oswald [2010], predicted ideal point personality scores very well from much simpler scoring methods). In these cases, there is no apparent empirical advantage for ideal point models applied to traditional personality measures. Perhaps this is fortunate, in the sense that the vast pool of personality test findings from past organizational research and meta-analyses do not need to be revised or reversed on the basis of scoring methods aligned with the dominance model.

The potential added value of ideal point models would come from those personality items that are situated in the middle of the trait continuum, such as an orderliness item that reads “Sometimes I am neat and tidy, but other times I am not.” We will refer to items like these as *moderate level*. Unlike items worded at the extremes, moderate-level items tend to be complex or multidimensional (e.g., see the attitude items in Roberts, Laughlin, & Wedell, 1999), yet they often rely on unidimensional Likert-scale responses. This is generally why complex items should be

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avoided (Hinkin, 1998), unless such items allow for similarly complex responses.

Returning to the example of orderliness, most people are not obsessively orderly nor are they incredibly disorganized. In fact, the very extremes of orderliness or any personality trait may reflect impaired life functioning or psychopathology (Markon, Krueger, & Watson, 2005). Therefore, test development efforts that are focused on moderate-level personality items might pay off in terms of higher acceptability or face validity as perceived by test takers. However, this advantage is not enough in and of itself to justify the switch to ideal point models. The use of moderate-level personality items and the ideal point model used to score them ought to yield a practical increase in reliability and validity when compared with simpler items scored by the dominance model. In other words, one wants to avoid using a more complicated measurement paradigm corresponding benefit trade-off. Science prefers parsimony unless the added complexity is justified (i.e., Occam's razor). One could predict that reliability and validity would be especially pronounced for people located in the middle of the trait continuum when utilizing the ideal point measurement model—a lot of people, given that most personality traits approximate a normal distribution. This prediction might bear itself out because high agreement on a moderate-level item indicates that a person has a moderate trait standing, whereas high disagreement means that a person could stand either high or low on the trait.

Therefore, it is worthwhile for future research to investigate the content and response processes for moderate-level items, given their potential to improve our measurement and understanding of personality traits. There is much work to be done, however, because there are at least three interpretations when a person strongly and positively endorses a moderate-level item such as the orderliness item "Sometimes I am neat and tidy, but other times I am not." One interpretation is straightforward

and follows the tenets of good personality measurement, where the person is assumed to possess a moderate level of the trait of orderliness, and the prediction of orderly thoughts, feelings, and behaviors is stable, albeit imperfect. A second interpretation is that the person possesses a highly nuanced version of orderliness, such that orderliness is highly contingent on psychological judgments or perceptions of the particular situational features at hand (e.g., see the CAPS model of Mischel & Shoda, 1995, or the trait-situation interactionist model of Tett & Burnett, 2003). The third interpretation is that the person is indifferent to orderliness, and any thoughts, feelings, and behaviors pertaining to orderliness are inconsistent and unpredictable. Critically, the second and third interpretations imply a very high or very low level of situational judgment or decision making (respectively), perhaps as much or more than they imply anything about a person's standing on a personality trait. To the extent these three possible interpretations are all viable ones, there is the potential for a high level of ambiguity from the typical moderate-level personality item. Advances in measurement content, delivery, and response processes may help disentangle this interpretational problem.

It is important to note that ambiguity not only exists for these moderate-level items that make the ideal point response model distinctive from the dominance model; it also applies to traditional Likert-scale items that are worded at the extremes. For Likert-scale items, however, the ambiguity shifts from the "middling" item content to the middle of the response scale. In fact, the research literature contains several empirical attempts to determine whether Likert-scale responses in the middle category are truly indicating a moderate level of endorsement of personality test items, as opposed to reflecting indifference or confusion about their content (Hernández, Drasgow, & González-Roma, 2004; Kulas & Stachowski, 2009; Rosenberg, Izard, & Hollander, 1955).

Issues in Need of Research

The Drasgow et al. paper and the present discussion together suggest several questions we view as important ones to research:

1. What is the practical impact when applying the ideal point versus dominance model approach to scoring measures of personality traits? Does it meaningfully change the magnitudes or patterns of validity for predicting organizational outcomes? Work on ideal point models has typically focused on reliability estimation, so the ongoing validity work mentioned by Drasgow et al. is very important because it can help to determine whether the validity of an innovative personality measurement system used in a high-stakes testing setting is driven by the forced-choice test format, by moderate-level personality items coupled with ideal point modeling, or both.
2. To improve the reliability and validity of people's trait scores in the middle of the trait continuum, is it better to add more moderate-level personality items (where the content is somewhat ambiguous) or more items at the high and low extremes (where a moderate-level response is somewhat ambiguous)? Impressively, Drasgow et al. found that over 25% of the items for their personality scale of order were better fit by the ideal point model because their moderate trait levels (Chernyshenko et al., 2007). It would also be valuable to know the size of the resulting increments in conditional reliability and validity along the trait continuum.
3. Does removing the middle category from a Likert scale reduce problems with responding to personality items with extreme wording, or does it merely disguise those problems or transform them into new ones?
4. How do subject matter expert judgments of item trait levels compare

with the item locations empirically estimated from personality item data scored with ideal point item response theory models? What is the meaning of any large discrepancies, and does it have implications for item development and scoring?

5. How might supervisor or peer rankings of employees on personality traits improve the reliability and validity of those situated in the middle of a trait continuum, compared with self-report on moderate-level items with ideal point scoring? Sociometric ratings by peers and supervisors are consistent with the important concept of personality as reputation (Hogan, 1996). An approach to obtaining such ratings can be found in a current theory-driven method in the context of performance appraisal (Goffin, Jelly, Powell, & Johnson, 2009).

Thurstone was the person who proposed the ideal point model, but he also proposed the idea of simple structure in factor analysis (Thurstone, 1935), and he would likely agree that we should stick with simpler models until the data suggest that model complexity from ideal point scoring will improve model fit (Pitt & Myung, 2002; Preacher, 2006). Drasgow et al. suggest that we are in fact headed in this latter direction, as the substance of personality test development catches up with the statistical models and computer technology that allow for much greater flexibility in the delivery and scoring of personality tests. In our view, there is some distance to go, but we are making good headway as we enter an exciting era of research and applications involving personality testing in organizations.

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Thurstone Might Have Been Right About Attitudes, but Drasgow, Chernyshenko, and Stark Fail to Make the Case for Personality

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Drasgow, Chernyshenko, and Stark's (2010) polemic argues the need for, and superiority of, an ideal point response process and an ideal point measurement model, for "personality" measurement, broadly defined. Yet, the target article leaves me unconvinced of their merits or necessity in personality trait measurement. The reasons for my skepticism are their (a) questionable distinction between cognitive and noncognitive data as it relates to the applicability and interpretability of dominance response models, (b) weak conceptual link between attitude measurement theory and personality trait assessment, (c) reliance on the findings produced from a single fit index when applied to a limited set of self-report measures, and (d) failure to provide empirical evidence that personality trait scales, in general, fail to include items that provide measurement precision in the "middle" or "intermediate" level of the trait range.

Cognitive and Noncognitive Measures: Different, but Not That Different

Throughout the target article, a line in the sand is drawn between personality (typical performance) and cognitive

(maximal performance) measures and resulting item response data. This distinction is necessary for their arguments because an ideal point response process is hard to fathom for cognitive abilities. Thus, the authors confine their arguments to self-report measures of "personality" broadly conceived. Repeatedly throughout the article, personality data and attempts to fit dominance models (e.g., factor analytic or item response theory) to them are referred to in derogatory terms, relative to the apparent shining armor of cognitive tests. For example, "We believe that dominance models are most sensibly applied to domains in which an individual's capacity or maximum performance capability is pitted against the difficulty or extremity of the item" (p. 467) and "... psychometric models for dominance response processes, such as classical test theory, factor analysis, and logistic item response theory models, are ill suited for response processes requiring introspection" (p. 467).

I sympathize with some of their arguments. There are important applied and conceptual differences between cognitive and noncognitive (personality, psychopathology, and patient reported outcomes) measures and resulting item response data (Reise, in press; Reise & Waller, 2003, 2009). For example, many personality and psychopathology constructs are more like unipolar "quasi-traits," definable at only one end of the continuum

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(e.g., depression). In turn, my position is that these differences need serious attention when considering the application of item response theory (IRT) measurement models. Moreover, I agree with the authors that the field of self-report personality assessment, relative to the large-scale cognitive assessment, is burdened by the weight of unexamined trait constructs (Tellegen, 1988, 1991), poorly constructed and psychometrically analyzed scales, score distorting or invalidating response artifacts including method effects, and research that continues with little regard for understanding the underlying response process or the meaning of the latent trait scale (Block, 2010; Borsboom, Mellenbergh, & van Heerden, 2003; Reise & Waller, 2009).

Nevertheless, Drasgow et al. take the cognitive versus noncognitive measurement distinction way too far. First, cognitive tests, such as an eighth grade social studies statewide achievement test, certainly involve an "introspection" component in the form of two fundamental cognitive processes—recall and recognition. Second, and more importantly, of all the well-acknowledged problems in "personality" measurement, there is no compelling evidence that either personality scales in general do not measure well in the "intermediate" range of the construct or that dominance response process models are inadequate. Sweeping statements, such as those cited above, cavalierly dismiss all the exploratory and confirmatory factor analytic research and applications of IRT models to measures of noncognitive constructs that have appeared in well-respected journals (as reviewed in Reise & Waller, 2009). It also ignores the fact that some of psychology's most heavily researched and most validated scales are personality or psychopathology measures developed under a dominance response framework.

Can Personality and Attitude Measurement Be Lumped Together?

The article begins with a famous quote "Attitudes can be measured," yet the

authors spend the remainder of the article trying to expand their arguments from attitude measurement to personality measurement more generally. This isn't surprising. Attitude measurement is a small and highly specialized field of limited interest to a broad audience, especially an applied audience; few individuals are hired, fired, or promoted because of their attitudes toward partial birth abortion, gun control, or the length of sentencing for property crimes. Hence, to justify a more general applicability of the ideal point response process, and the associated psychometric model, the authors link personality measurement with attitude measurement. They accomplish this by arguing "personality is an attitude about the self" because personality self-report items require "introspection."

I cannot think of a single personality theorist who would agree with this theory. Consider the items of the California Q-sort (Block, 1961), which taken as a whole is a pretty good definition of the personality domain. Is a tendency toward undercontrol an attitude about the self? How about a high energy level or difficulties handling stress? Moreover, the notion that personality is *merely* "an attitude about the self" does not appear consistent with the empirical evidence that (a) personality measurements of the same construct, either through self-reports, other reports, or behavior tasks, tend to be highly correlated (Furr, Wagerman, & Funder, 2010; Nave, Sherman, & Funder, 2008); (b) personality measures have proven heritabilities (Stubbe, Poshuma, Boomsma, & De Geus, 2005); (c) personality measures are related systematically to biological parameters, such as those assessed via fMRI (Eisenberger & Lieberman, 2004); (d) personality traits are resistant to change and consistent over long periods of time (Friedman et al., 1995); and, most importantly, (e) personality measures correlate meaningfully with important life outcomes (Ozer & Benet-Martinez, 2006).

These empirical facts are more consistent with a view of personality as a stable, biologically based causative mechanism

as opposed to reflecting “attitudes about the self.” For example, “We can begin by defining a *trait* as an inferred relatively enduring organismic (psychological, psychobiological) *structure* underlying an extended family of behavioral dispositions. In the case of personality traits it is expected that the manifestations of these dispositions can substantially affect a person’s life” (Tellegen, 1991, p. 13). In fact, if one adopts a classic Allportian view of personality traits as being determinative (i.e., influencing a person’s perception of a situation), it appears that personality operates to influence a person’s introspection processes and thus their attitudes about the self.

If the authors intended to limit their arguments to the cognitive process of responding to self-report personality items, rather than to argue that attitudes and personality traits are essentially the same, I would be more sympathetic to their perspective. Yet I find it puzzling that not a single focus group, case study, or “talk out loud while responding to items” research study was cited that might lend some empirical support that, at least for some self-report personality items, the dominance response does not hold. It seems to me that such fundamental research is necessary and would make an important contribution to personality assessment theory.

Are Ideal Point Models Empirically Supported?

Drasgow et al. argue that their research demonstrates that personality data do not fit dominance IRT models well. Moreover, they argue that their research demonstrates that personality measures fail to include items of “intermediate” difficulty (i.e., provide discrimination in the middle of the trait range), and when such items are written and then fit to ideal point models, such items reveal discrimination ability. In short, the so-called “intermediate” items are not necessarily bad, it is just that psychometricians haven’t been fitting the “right” or “superior” model.

Yet the “evidence” to support the need for ideal point models is based on the use of a single chi-square badness-of-fit test and their own rules of thumb for defining “excellent,” “satisfactory,” and “misfit.” More specifically, their studies on the sixteen personality factor and another self-created measure showed that item pairs and triplets displayed a relatively poor fit (i.e., high ratio of chi-square to degrees of freedom). It is not at all clear whether their index provides any meaningful gauge of how wrong the estimated IRT item parameters were or what the applied consequence of such misfit was. It is also not clear whether the misfit was attributable to poor scale construction. For example, personality scales are notorious for containing repeated item content, which, in turn, causes violations of local independence. Such local dependence would almost certainly cause a large chi-square for item triplets or pairs.

It is a leap of faith to argue that badness-of-fit, as judged by a chi-square test, indicates an ideal response process; it may just as well have been generated by multidimensionality, or simply reflect poor item construction. How those competing explanations are evaluated is not clear in the target article. At least dominance response models propose a common latent variable that is “causal” of item variance and thus inter-item covariance (Bollen & Lennox, 1991). Moreover, dominance models allow researchers to make informed judgments as to which items belong to which constructs and which items are “bad.” Making such determinations is a complicated and time-consuming process if done seriously (Tellegen & Waller, 2008). I am not convinced that ideal point models afford such empirically based judgments. The examples of items provided, “I enjoy chatting quietly with a friend at a café” (Extraversion) and “My life has had about an equal share of ups and downs” (well-being) leave me questioning whether the proposed underlying latent variables can sensibly explain item responses.

Finally, in support of future measures based on ideal point models, the

authors claim "Some constructs in organizational psychology might be better studied by embracing an ideal point perspective" (p. 472). They then mention person–organization fit, employee performance, organizational commitment, leader behavior, and perceived organizational support. I question whether any latent trait model, including ideal point models, can be usefully applied to these types of socially constructed emergent variables (Bollen & Lennox, 1991). On the other hand, I found the suggestion of reviving forced-choice personality questionnaires at least intriguing, and I am anxious to see if such measures can add anything beyond existing competing measures such as the California Q-set (Block, 1961).

Conclusion

Although I have disputed many of their iconoclastic claims, I agree with the authors on two critical issues. First, it is important to get the psychometric model right, and research should continue that considers the question of what is the right model for specific types of personality measures. Second, we apparently agree that it is critically important to understand the item response process and the nature of the latent variable in personality measures. How are researchers to understand change when they do not understand the processes underlying the items? To the extent that the Drasgow et al. article motivates research along these lines, it has served a useful purpose. To the extent that it motivates researchers to create more redundant and unnecessary self-report "personality" scales, it is a distraction.

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Issues That Should Not Be Overlooked in the Dominance Versus Ideal Point Controversy

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Drasgow, Chernyshenko, and Stark (2010) argue that “responses to questions requiring introspection involve a comparison process,” and we agree with this statement. The key question is, however, what do respondents compare themselves to? Drasgow et al. suggest that respondents *always* compare their self-perception to the statement’s location (an ideal point); therefore, ideal point models should always be used. We feel the answer depends on the construct being measured and on the type of items used to measure it. Sometimes the response mechanism appears to be an ideal point process. But many other times we believe that given a statement and a binary (endorse/not endorse) rating scale, an individual will endorse the item if its utility is larger than a certain threshold and will not endorse it otherwise. In this case respondents compare themselves to a threshold, which calls for the use of a dominance model.

To investigate whether a threshold or an ideal point mechanism best describes the response process, intermediate items should

be used more frequently. This is because, as Drasgow et al. point out, if only extreme items are used, dominance and ideal point models will yield a similar fit. It is only in considering intermediate items like “My life has had about equal amounts of ups and downs” that we can tell apart an ideal point and a threshold process. However, classical statistics (i.e., low item–total correlations) is not the only reason why there are very few intermediate items in applications. There are other important issues with ideal point items and models that are overlooked in Drasgow et al.’s account. We review several of these issues below.

Good Intermediate Items Are Difficult to Write

Good item writing practice should be maintained, regardless of whether an item is extreme or intermediate, and regardless of whether it reflects a dominance or an ideal point process. Items should be fair and should not confuse respondents. However, following this basic practice makes it difficult to write good intermediate items. To illustrate, consider descriptions of typical behaviors of an average scorer on a personality scale, as often given in questionnaire manuals or in feedback reports. These descriptions are very fuzzy, with suggestions that some behaviors might apply in certain conditions, or in certain

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times, or that behaviors might apply to a lesser extent. This ambiguity very much applies to items with locations in the middle of the trait continuum.

The first problem occurs when some conditional clause is used to describe an average scorer, which leads to double-barreled items. Consider the item "Although I have a daily organizer, I have a hard time keeping it up to date." Now consider a respondent trying to make sense of this item and deciding how to respond. The typical feedback is that such items are incredibly frustrating. To some, the first part does not apply and it makes the whole item meaningless, thus considerably increasing the likelihood of responding randomly. To others, the first part applies but the second part does not apply, and they feel equally confused. We believe that regardless of the underlying response model, most of these items are to be thrown away, not because of low item-total correlations, as we advocate the use of item response theory (IRT) modeling instead, but because we are concerned about the face validity of such items and potentially about their construct validity too.

The second problem occurs when items explicitly require a judgment about how one compares with a reference group. An example of such an item is "My room neatness is about average." This item requires making a judgment about the state of other people's rooms. Such judgments are confounded by the respondent's frame of reference and tend to show item bias. This situation can occur also with extreme items such as "I am more outgoing than most people." Such items tend to cause problems when people with very different frames of reference are compared, for instance in cross-cultural research. Consequently, outgoing Spaniards might score lower on sociability than, let's say, Finns.

The third problem occurs when an attempt to write an intermediate item leads to introducing particulars and contexts not clearly related to the construct of interest. Consider the item "I enjoy chatting quietly with a friend at a café" (as opposed to

simply "I enjoy chatting"). This item is intended to measure Extraversion; however, it is likely that the specificity of the context will introduce multidimensionality, and the item will have a low discrimination on the Extraversion scale.

Developers of personality scales know that it is easy to write descriptions of high and low scorers on a personality trait. Because dominance items describe behaviors typical of the high or low scorers on the trait, they are also easy to write. They are easy to answer, too. From a respondent's perspective, responding to the item "I am organized" is straightforward. This item is unambiguous and is easily related to one's behavior. The fact that it is easier to write good dominance items than good ideal point intermediate items is an overlooked issue in Drasgow et al.'s account.

Estimation May Not Be as Accurate for Ideal Point Models

An IRT model is useless unless it can be shown that its item characteristic curves (ICC) can be estimated with enough precision in reasonably small samples and that latent trait estimation is precise enough. Having been used more widely to date, more software is capable of fitting dominance models, and requirements for successful item and latent trait estimation are better known than for ideal point models. Software considerations aside, it may well be that it is inherently more difficult to recover true item parameters for ideal point models than for dominance models. Maydeu-Olivares, Hernández, and McDonald (2006) proposed the ideal point counterpart of the normal ogive model, the normal probability density function model. Yet, in simulation studies it was not possible to obtain nearly as accurate an estimation of item parameters as with the normal ogive model (Forero & Maydeu-Olivares, 2009). Clearly, more research involving simulation studies is needed with ideal point models to show that item parameters, as well as respondent's traits, are estimated as well as they are with dominance models.

Ideal Point Models Are Not Invariant to Reverse Scoring

Within an IRT framework, it is not necessary to reverse score items, regardless of whether a dominance model or an ideal point model is used. However, some pause is needed when choosing the direction of the construct using ideal point models. In these models, reverse scoring of the items leads to a different set of parameters and to a different goodness of fit. See Maydeu-Olivares et al. (2006, p. 467) for an example involving modeling dissatisfaction with life as opposed to satisfaction with life. Dominance models, on the other hand, are invariant to reverse coding and allow equivalent modeling of either end of psychological constructs. That is, when dominance models are used and some or all items are reverse scored, parameters are transformed in a simple way, and model fit remains unchanged. This is not the case with ideal point models.

IRT Modeling of Forced Choice Does Not Require Ideal Point Items nor Models

Some researchers mistakenly believe that forced-choice measurement using IRT requires ideal point items and models. The confusion probably stems from the fact that the same group of researchers happened to extensively use ideal point models and items (Chernyshenko, Stark, Drasgow, & Roberts, 2007; Stark, Chernyshenko, Drasgow, & Williams, 2006) and have introduced an IRT approach to creating forced-choice tests that involve an ideal point model (Stark, Chernyshenko & Drasgow, 2005).

Modeling forced-choice responses requires a model for the response mechanism by which respondents choose between two items. The model for this response mechanism is independent of the model used to link an item to a personality trait. The latter can be a dominance model or an ideal point model.

Stark et al. (2005) used the multi-unidimensional pairwise preference model

(MUPP) for forced-choice items, in which the probability of preferring one item to another is approximated by the joint probability of endorsing one item and rejecting the other. To link the probability of endorsing the item to a personality trait, they use an ideal point model. However, there is nothing in the actual MUPP model that stops it from being populated with dominance items and, consequently, using a dominance model.

The MUPP is not the only model using an IRT approach to modeling forced-choice responses. Our own model (the Thurstonian IRT model, see Brown & Maydeu-Olivares, in press) uses Thurstone's (1927) Law of Comparative Judgment to model how respondents choose between two items. The model posits that respondents choose the item with the largest utility at the time of comparison. How the utilities of individual items depend on the psychological constructs being measured is immaterial for the comparative law. An assertion by Drasgow et al. that using a dominance model to describe forced-choice judgments can lead to serious model misspecification (see footnote in focal article, pg 473.) is misleading. We happen to use dominance items and a dominance model, but ideal point items and model could be used just as well. The problem, as Drasgow et al. rightly point out, is that when scored traditionally these instruments produce ipsative data. However, it is not the use of dominance items that produces ipsative data; it is the classical way of scoring forced-choice instruments. The Thurstonian IRT model we proposed completely overcomes the problems of ipsative data in questionnaires with dominance items and ICC, and latent traits are very accurately recovered (Brown & Maydeu-Olivares, in press).

Furthermore, Tsai and Böckenholt (2001) have shown that ideal point and dominance models are indistinguishable from comparative data. That is, from preference choices made between two items it is impossible to determine whether the items were linked to personality traits using an ideal point model or a dominance model. Hence, it

does not matter whether ideal point or dominance items are used in forced-choice formats. Clearly, much additional work is needed to evaluate the implications of different item response models on comparative judgments.

Dominance Items Measuring Multiple Traits Can Be Mistaken for Ideal Point Items

In applications, even the best constructed unidimensional measures contain sizeable amounts of multidimensionality. Drasgow et al. report that their interest in ideal point models arose when fitting unidimensional dominance models to personality scales and finding that dominance models could not reproduce personality data well. Searching for better fitting models, they applied a nonparametric unidimensional model (Levine, 1994), which yielded shapes that suggested the use of ideal point models.

Unfortunately, their findings should not be taken to imply that an ideal point model is the “true generating” model for the data. Rather, if the true model is a multidimensional dominance model, fitting Levine’s model would reveal bumps and swirls similar to those for a unidimensional ideal point model (Levine, 1994; Maydeu-Olivares, 2005). Although Drasgow et al. argue that some debates about personality dimensions “might be clouded by the application of misspecified models” (implying that multidimensionality might be wrongly suspected where a unidimensional ideal point process is present), it is equally probable that the opposite can occur (i.e., a unidimensional ideal point process is wrongly suspected where a multidimensional dominance model is present; see Peress & Spirling, 2009). The presence of multidimensionality, however, might not be picked up if the construct is tested in isolation from other constructs, and no other items share variance specific to this other dimension. For instance, it might be impossible to tell from the data if the item “I enjoy chatting quietly with a friend at a café” is an ideal point item on the one-dimensional extraversion scale or a dominance item

measuring introversion and warmth. This admittedly messy situation suggests that one cannot differentiate between the two models on the basis of model fit alone, and we agree with Drasgow et al. that a good theory should guide judgments made about appropriateness of any model.

Conclusions

It is one thing to find that *some* items are better described (importantly, not only mathematically but also conceptually) by the ideal point response process. In this case, it is indeed a welcome development to model such items appropriately. However, it is an entirely different thing to contend that applied researchers and practitioners should write *all* their items to fit an ideal point response model. We feel that there are clear advantages in using dominance items (and models) that should not be overlooked.

Our own approach regarding the choice of dominance versus ideal point models is simple. We inspect the item stems and classify items as ideal point or dominance items. We then fit a dominance model if all items are dominance items and an ideal point model if all items are ideal point items. We never limit ourselves to unidimensional models. Rather, we always fit multidimensional models as well. In the case of items of mixed type, we first fit an ideal point model. This is because one can argue that many dominance items are simply a special case of ideal point items where the ideal point is located outside the region of high density of respondents. However, given the issues discussed above, we also fit a dominance model of the same class for comparison. Clearly, most items in use today are dominance items. Thus, we end up fitting more dominance models than ideal point models. Even when ideal point items are present, sometimes we end up choosing a dominance model for one or more of the reasons listed above.

Should we use ideal point models for all items? Our position is a clear “no,” and we have offered a number of arguments to support this. Should we discard existing

dominance items and replace them with ideal point items? If dominance items work well and provide good measurement quality and validity, as they do in personality measures, our answer is "certainly not." Should we develop more intermediate (including ideal point) items? "Of course." Only by examining more intermediate items, in more situations, can we obtain clear answers about their properties, advantages, and disadvantages, and resolve the ideal versus dominance models controversy.

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Two Caveats for the Use of Ideal Point Items: Discrepancies and Bivariate Constructs

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Ideal point items in general and the specific item response theory-based models outlined by Drasgow, Chernyshenko, and Stark (2010) hold considerable promise for the measurement of intermediate levels of constructs such as beliefs, traits, and attitudes. The aim of this commentary is to outline two caveats relating to the use of a certain type of ideal point item: one related to the assessment of discrepancies (e.g., person–environment fit) using ideal point items and one related specifically to possible computer adaptive testing applications. Ideal point items for attitude or trait measures as outlined by Drasgow et al. appear to take one of the two forms. The first (assuming Likert items) is a statement of the general form “I have moderate levels of construct X,” whereas the second takes the general form of “I have equal amounts of A and B” or “I am characterized by a balance between A and B,” where A and B represent the two endpoints of a bipolar scale. This commentary focuses on this second possible form for an ideal point item, hereafter referred to as a double-barrel ideal point item.

Assessing Discrepancies

The first concern discussed in this commentary relates to the possible use of ideal point items to assess discrepancies such as might occur when assessing person–organization fit. Drasgow et al. argue that ideal point items might be written that directly capture the difference between the organizational environment and the employee’s needs. Edwards (2001) has previously highlighted the various problems associated with difference scores in general and with the specific approach that entails asking respondents to mentally compute difference scores in order to formulate responses to questions of the type described by Drasgow et al. Readers are referred to Edwards’ (2001) paper for a detailed explanation, but a key problem with this approach is that respondents who are asked to use the perceived difference between two constructs (e.g., actual amount of autonomy and wanted amount of autonomy) in formulating their responses to these types of items do not appear to base their computation of the difference between two constructs on the actual components of the difference. Edwards (2001), for example, reanalyzed data from Rice, McFarlin, and Bennett (1989) and was able to show that the perceived difference between what the job provided and what was desired (the difference judgment) was only moderately related to the difference between judgments of what was provided and what was desired

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(the two component pieces). This suggests that ideal point items asking respondents to base their responses on perceived differences may be assessing something different than the actual perceived difference.

Bivariate Constructs

The second concern relates to the possible use of double-barrel ideal point items in the construction of computer adaptive tests for traits and attitudes. The vast majority of psychological constructs are bipolar in nature such that each individual can be described by their position along a single continuum ranging from low levels of the construct to high levels of the construct. There are, however, important constructs that may be *adequately* represented by a bipolar structure but where significant theoretical and practical utility is gained by a measurement approach that acknowledges their bivariate structure. That is, an individual's standing on the construct may best be plotted on a two-dimensional plane rather than on a single axis. Ideal point items that take the second form outlined above make it difficult to accurately assess an individual's position on such a two-dimensional plane. Three simple examples serve to illustrate this issue.

First, attitudes are typically thought to be bipolar in nature with individuals ranging from negative attitudes toward an object to positive attitudes toward the object. This conceptualization of attitudes is appealingly parsimonious, but there is evidence that important information is retained if an individual's attitude is described by its position along two dimensions, a positive attitude dimension and a negative attitude dimension (The Bivariate Evaluation Plane; Cacioppo & Berntson, 1994; Cacioppo, Gardner, & Berntson, 1997, 1999). This bivariate approach to the measurement of attitudes allows one to distinguish between those individuals who simultaneously have high positive and high negative attitudes toward an object or behavior (ambivalence) and those individuals who simultaneously have low positive and low negative

attitudes (indifference). Such a distinction is not possible using a bipolar view of attitudes whereby each of these individuals is viewed as having intermediate (i.e., balanced) levels of the attitude. Interestingly, the bivariate approach to attitudes has been shown to increase the ability to predict attitude-relevant behaviors. For example, individuals with ambivalent attitudes toward blood or organ donation have been shown to be more likely to donate blood or organs (Cacioppo & Gardner, 1993) than those with indifferent attitudes, and recent work on job attitudes (Credé, Chernyshenko, Bagraim, & Sully, 2009) suggests a similar effect for the relationship between job attitudes and workplace behaviors. A hypothetical ideal point item such as "I experience as much satisfaction in my job as I do dissatisfaction" is likely to exhibit ideal point characteristics if job satisfaction is viewed as a bipolar construct, but would not only require respondents to make the problematic type of difference score computations discussed by Edwards (2001) but would also obscure the difference between an employee who experiences neither satisfaction nor dissatisfaction (i.e., indifference) and an employee who experiences both satisfaction and dissatisfaction (i.e., ambivalence).

Second, psychological well-being can be viewed as a bipolar construct with individuals ranging from low levels of well-being to high levels of well-being or as a bivariate construct with individuals being characterized by differing combinations of low well-being and high well-being. Ideal point items of the kind being discussed in this commentary may obscure this bivariate distinction. Consider, for example, the item used by Drasgow et al., as an example of an ideal point item measuring well-being: "My life has had about an equal share of ups and downs." This item is likely to be good for assessing intermediate levels of a bipolar well-being construct, but it appears to equate individuals with no ups and downs (at a clinical level a possible indicator of depression) with an individual with high levels of both ups and downs (at a clinical

level a possible indicator of bipolar disorder). There may, of course, be contexts in which a bipolar approach to measure well-being is appropriate, but we would urge researchers to carefully consider the possible loss of information inherent in such an approach. If, for example, the setting is one in which the distinction between high ups and high downs versus low ups and low downs is meaningful and important, then ideal point items of the first kind (e.g., "I experience a moderate amount of ups in my life" and "I experience a moderate amount of downs in my life") may be preferable to a double-barrel ideal point item.

Third, gender identity can be thought of as falling along a single continuum ranging from maximally feminine to maximally masculine, but many researchers consider these two endpoints to represent distinct dimensions (Bem, 1977; Spence & Helmreich 1978), allowing them to distinguish between individuals who describe themselves as possessing high levels of both characteristics (androgyny) and individuals describing themselves as having low levels of both characteristics (undifferentiated). This distinction appears to be important for understanding psychological well-being (Jones, Bloys, & Wood, 1990), but a hypothetical ideal point item such as "I consider myself to be just as feminine as I consider myself to be masculine" would obscure this distinction.

The bivariate structure of some constructs represents a special challenge for the construction of computer adaptive tests (CATs) using double-barrel ideal point items. Consider three possible items in a CAT of job attitudes: "There is much that I love about my job," "There is much that I hate about my job," and "There is as much that I like about my job as I dislike about my job." The first two of these are likely dominance items, whereas the third is likely an ideal point item. An employee with ambivalent job attitudes (i.e., high positive and high negative attitudes) would be likely to endorse all three of these items but is unlikely to be presented with all three in a CAT because an endorsement of any one

of these would likely preclude the other two from being presented to the respondent. As a result the ambivalent nature of that employee's job attitudes may not be correctly assessed. Similarly, a CAT of job attitudes comprised mainly of double-barrel ideal point items is likely to lead to employees with ambivalent attitudes being equated with employees with indifferent attitudes.

Conclusion

The use of double-barrel ideal point items should be approached with caution in two narrow domains: the assessment of discrepancies and the assessment of an individual's standing on bivariate constructs using CATs. In general, ideal point items do not appear to pose a threat to the assessment of an individual's standing on bivariate constructs as long as test developers are aware of the possible bivariate nature of constructs and construct items in a manner that reflects this possibility. For example, ideal point items could easily be constructed to assess intermediate levels of a negative attitude and intermediate levels of a positive attitude thereby allowing researchers to accurately position an individual's position on the bivariate plane. Some of the ideal point items discussed by Drasgow et al.—double-barrel items—would, however, potentially obscure the bivariate nature of some constructs when used in CATs.

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Six Questions That Practitioners (Might) Have About Ideal Point Response Process Items

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The ideal point model discussed in Drasgow, Chernyshenko, and Stark (2010) has the potential to advance the understanding of item responses, and the method has significant implications for psychometric practice. Although Drasgow et al. outlined some avenues for applying the results of the ideal point response process, questions and challenges about applying these models in real-world psychometric settings remain. The goal of this commentary is to present questions that industrial–organizational (I–O) practitioners may have about using ideal point measurement in their work. These six questions highlight gaps in our current knowledge that

will require further research to adequately address.

Question #1: How Can an Ideal Point Measure Be Scored Without IRT-Based Parameters?

The original impetus for Likert (1932) to develop his scoring method was the difficulty associated with scoring Thurstone's (1928) original measures. Thurstone's method also required a large calibration sample to scale the items on the latent trait distribution. Likert's method of scoring was simpler to create and use (e.g., simply adding scores) and did not require a large sample size. Later, advances in statistical techniques like Roberts and Laughlin's (1996) development of an item response theory (IRT) model to analyze ideal point measures helped solve some of the issues facing researchers like Likert and Thurstone. Applying these IRT models to ideal point measures, researchers can estimate respondents' standing on the

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latent trait continuum. However, applying these IRT models still requires large sample sizes, and assessing unidimensionality with these models is difficult and understudied. Drasgow et al.'s recommendation of calculating the mean of item locations also requires a calibration sample that an I-O practitioner may not have. For practitioners operating on a global scale, this presents an even more severe issue as multiple calibration samples would be required to ensure a single scoring algorithm or to develop culturally based scoring algorithms.

Given these limitations to existing scoring methods, the first question we believe a practitioner may have is "How do we score ideal point measures without the use of calibration samples or IRT?" That is, if I-O practitioners have an N of 100, how can we score an ideal point measure? With traditional Likert scoring systems, more is always better—except for reverse-coded items—with more positive options receiving higher scale values. With ideal point items, the relationship between scale options and scale values may be nonlinear. Rubrics for understanding these scoring schemes will be essential, especially for practitioners who do not have access to, or the capability for, running IRT models to score respondents.

Question #2: What Should Practitioners Do If They Want to Create an Ideal Point Scale, but They Only Have a Small Sample Size?

Beyond requiring large sample sizes to score ideal point measures, practitioners need large sample sizes to develop and evaluate ideal point scales. Applying IRT models to scales requires sample sizes large enough to obtain stable parameter estimates. As the number of item parameters within a model increases, the sample size required to estimate those parameters with reasonable precision increases. A practitioner will need a larger sample size to fit

an ideal point IRT model, as opposed to a dominance model. Roberts, Donoghue, and Laughlin (2000) suggest that a minimum of 750 participants are needed to uncover stable parameter estimates for a 20-item measure, making it difficult to apply the generalized graded unfolding model (GGUM) to measures. Even if one chooses to utilize Thurstone's method of scaling items, a sample size of nearly 300 is needed (per Thurstone's [1928] recommendation). Regardless of whether practitioners wish to use IRT modeling or Thurstone's original scaling method, they must obtain a large sample—likely larger than the applicant pool at their disposal—to evaluate an ideal point measure. Similar to scoring ideal point measures with a limited sample size, practitioners would benefit from any research or advice on creating and evaluating ideal point scales when only a limited sample size is available.

Question #3: How Do We Write Ideal Point Items?

Although the increase in precision that ideal point measures provide is positive, guidelines for creating ideal point items have yet to be developed. Indeed, Chernyshenko, Stark, Drasgow, and Roberts (2007) noted that writing ideal point items is difficult because (a) no classical test theory tools exist to evaluate quality of items and (b) Thurstone's original scaling method (discussed above) was cumbersome (p. 92).

Guidelines need to be written to provide item writers a taxonomy of discriminating qualities that produce items with significant unfolding. Many of the item-writing heuristics taught in tests and measures courses are based on the dominance model. For example, the heuristic of avoiding double-barreled items may no longer be appropriate for ideal point items.

In addition to guidelines on how to write good unfolding items, researchers will need to change the current method of evaluating items. Researchers who develop scales with dominance theory assumptions have the

luxury of using the tools of classical test theory (CTT). CTT provides researchers with guidance on good and bad items in small pilot samples, a practice with no analogue for unfolding models.

Question #4: What Are the Implications for Scale Length Using Ideal Point Assumptions to Create Scales?

Related to writing effective ideal point items, little information exists about the requirements for scale length created with ideal point items to achieve reasonable estimates of standing on the latent trait. Similar issues regarding scale length have been debated in the literature related to dominance items, and some guidelines have been suggested based on CTT. For example, it has been suggested that as few as three items per scale can be used to obtain an adequate level of internal consistency (see Hinkin, 1998 for review), or a minimum of three to four items are needed to adequately scale a latent construct in structural equation modeling (Kline, 2005). Finally, one can estimate the number of (parallel) items needed to obtain a level of reliability by applying the Spearman–Brown formula (Allen & Yen, 1979).

Ideal point measures need to be longer because of the need to untangle responses from people throughout the range of the trait distribution. Respondents who disagree on a series of items may be either extremely low or extremely high on a particular trait; it is only in conjunction with more moderate items that these respondents can be differentiated. Given recent trends in industry for creating shorter tests to address concerns from candidates and business partners (e.g., recruiters), the creation of longer selection instruments or employee surveys could be problematic. Therefore, like advice on writing items, both I–O practitioners and researchers would benefit from greater understanding of how using ideal point assumptions to create scales will affect scale length.

Question #5: In the Noncognitive Area, Would a Practitioner Expect to See Higher Criterion-Related Validity Coefficients Through the Adoption of Ideal Point Measures?

Drasgow et al. mention that the adoption of ideal point approaches to the measurement of noncognitive domains results in better measurement precision of that construct. However, an important question is whether this increase in measurement precision results in stronger relationships between constructs. Given how resource intensive it is to create ideal point measures (e.g., large sample size and more difficult item writing), practitioners should be assured that their investments will yield returns. Therefore, researchers should look to establish that ideal point measure estimates of constructs are more strongly related to criteria than a simple sum score of dominance items. In one example, Chernyshenko et al. (2007) showed that their ideal point measure of orderliness had stronger criterion-related validities than did dominance measures.

When evaluating the relation of ideal point measures to external criteria, one must remember that validity coefficients are sensitive to the whole range of trait distribution. Therefore, it is possible that the correlation may change a little compared to dominance measures, but top-down selection decisions may differ because of the significant changes at the top end of the distribution. Therefore, researchers should not just focus on correlation coefficients when examining how ideal point models influence hiring decisions. I–O research can further inform I–O practice about the changes in validity associated with using ideal point measures to determine whether the benefits of using ideal point measures outweigh the costs associated with creating one.

Question #6: What Is the Applicability of Using the Ideal Point Model to Develop Measures of Engagement and Satisfaction?

Employee satisfaction and engagement are two constructs that are very popular in

applied and academic I–O work. Often measures of these constructs were developed using dominance models, making it difficult to interpret the midpoint of the scale. In other words, if a respondent is scored as “neutral” on an engagement scale, does this mean that this employee is moderately engaged or that she is ambivalent in her engagement? The ideal point response process of comparing one’s standing on a latent distribution of the construct to the location of the item on the latent distribution—what Drasgow et al. referred to as introspection—seems to fit the measurement of engagement and satisfaction better than the dominance approach. Preliminary evidence of this response process underlying satisfaction surveys may be obtained from Carter and Dalal (2010), showing that the Job Descriptive Index Work scale contained ideal point items.

If ideal point models measure these constructs better, it is possible that new questions about these constructs might be asked. For example, practitioners would no longer have trouble interpreting “neutral” scores. In addition, applying ideal point assumptions to scales of these constructs might yield more precise measurement. With an increase in measurement precision, I–O practitioners gain a better understanding of the employees in organizations and take action to improve their lives.

Conclusions

The study of ideal point response processes could still be considered in its preliminary stages. Although posited by Thurstone (1928) over 80 years ago, Likert’s (1932) more straightforward and less labor-intensive approach to measurement dominated. With the resurgence of interest in ideal point response models and new approaches to analyzing these measures

(e.g., GGUM), it seems reasonable to suspect that ideal point measures will see favor in the future. How long in the future is a matter of showing researchers and practitioners the value of this method. Indeed, Drasgow et al. lamented that it could possibly take as long as 30 years to see the widespread adoption of ideal point measures. The research on ideal point models is exciting and, as Drasgow et al. show, will likely revolutionize measurement practice and theory. We hope the questions we identified will be addressed by researchers as we move forward in our understanding of the ideal point response process.

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If Thurstone Was Right, What Happens When We Factor Analyze Likert Scales?

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Drasgow, Chernyshenko, and Stark (2010) rightly explain how the unfolding model can be a better representation of people's responses to what they term introspective judgments (e.g., ratings of attitude or personality) than the dominance model. Drasgow et al. noted Davison's (1977) argument that a set of unidimensional items would form two factors when factor analyzed if those items fit an unfolding model. Drasgow et al. further noted that the veracity of arguments about the underlying structure of a set of items depends on getting the response model right. We elaborated on these points in Spector, Van Katwyk, Brannick, and Chen (1997), and produced what we believe is an empirical example of the artifactual two-factor phenomenon using job satisfaction data. In this commentary, we further discuss the issue of artifactual factors produced by respondents disagreeing with oppositely worded items and provide data that this phenomenon can occur in a sizeable proportion of a sample.

Many summated rating scales use bipolar agreement responses in which respondents can indicate varying levels of agreement or disagreement. Items are often worded in both directions, such that agreement with positively worded items will represent high

values on the underlying construct, whereas agreement with negatively worded items represent low values on the underlying construct. It is assumed that responses to such items are opposite, such that respondents who moderately agree with items worded in one direction will moderately disagree with items worded in the opposite direction. As Spector et al. (1997) demonstrated with job satisfaction data, this is not necessarily the case. Drasgow et al. rightly noted that when the scale values of items are far from the respondent's standing on a trait, he or she will likely disagree with it, even if the item is on the same side of neutrality as the respondent. Thus individuals who are somewhat satisfied with a job will disagree with an item that is too positive, and a person who is somewhat dissatisfied with a job will disagree with an item that is too negative. This tendency to disagree with items written in both directions can affect inter-item correlations within a scale, which can result in a factor analysis result that encourages erroneous interpretation. The pattern of disagreement with both positive and negative items will produce a factor structure in which items load onto factors corresponding to item direction (see Spector et al., 1997 for a more complete explanation). In our experience such factors can be extremely clean with strong loadings on one factor and near-zero loadings on the other.

Do respondents actually disagree with items written in opposite directions? Yes, they do. We present an example from data ($n = 1,180$) on the Work Locus of Control

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Scale (WLCS; Spector, 1988), taken from Spector et al. (2001). We chose this scale because it assesses personality, which is the main focus of Drasgow et al.'s arguments and produces two very strong factors corresponding to items written in opposite directions (Macan, Trusty, & Trimble, 1996). The WLCS has 16 items, half of which are written in each direction—externality (the belief that the individual does not control rewards at work) and internality (the belief the individual has control over rewards at work). A 6-point agreement scale is used, with three levels of agreement and disagreement (slightly, moderately, and very much). To simplify the analysis concerning uniformity in response to oppositely worded items, we collapsed the level of agreement/disagreement into one level each, which were coded 1 for agree and 0 for disagree. We computed two-way contingency tables of responses (agree versus disagree) to each possible pair of the 8 externally written versus 8 internally written items, for a total of 64 possibilities.

We found that there was considerable variability across items in the percentage of sample who disagreed with items written in opposite directions, ranging from 5.7% to 35.6%, with a mean of 18%. For example, 35.6% of the sample disagreed with both the internal item "A job is what you make of it" and the external item "Getting the job you want is mostly a matter of luck." More than a quarter (26.9%) of the sample disagreed with both the internal item "Promotions are given to employees who perform well on the job" and the external item "To make a lot of money you have to know the right people." These results suggest that, at least for the WLCS, disagreement with items written in both directions was far from rare.

Shifting focus for a moment, Drasgow et al. noted that correlations (e.g., validity coefficients) will not be much affected by switching from a dominance model to an unfolding model. One point they did not mention was that for some jobs it may

be better to use an ideal point strategy in selection by matching the level of a characteristic between applicants and jobs. For example, some jobs may require an ideal amount of Agreeableness, Conscientiousness, or Extraversion (consider the jobs of police officer, first-level manager, and park ranger, respectively). In such cases, increases in utility might be possible by matching applicants to jobs based on minimizing the distance from the ideal point rather than maximizing the level of the applicant's trait.

Drasgow et al. have raised an important issue that can have important implications for both research and practice. At the current time, our knowledge of the extent of the problem is rather limited as generations of researchers and practitioners alike have assumed that the dominance model is adequate. Although it has served us well in many ways, it is time to give Thurstone a second look to be sure that we have not been led astray by uncritically accepting the dominance model in situations where unfolding does a better job of representing how people respond to our scale items.

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Cognitive Processes Related to Forced-Choice, Ideal Point Responses: Drasgow, Chernyshenko, and Stark Got It Right!

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The purpose of this reply to the Drasgow, Chernyshenko, and Stark (2010) focal article is to briefly explore why the ideal point model may prove to be superior to the dominance model for “introspective judgments” in the Drasgow et al. parlance. To do this, I will primarily use performance appraisal judgments as an example because this domain is where our main experience lies (Borman et al., 2001) although we have also used this approach in personality testing (Borman, Houston, Schneider, & Ferstl, 2005; Houston et al., 2003).

As noted by Drasgow et al., we developed computerized adaptive rating scales (CARS) to measure our three dimensions of citizenship or contextual performance using an approach similar to that described in Drasgow et al. In fact, the algorithm we used to iteratively select item pairs was developed by Stark and Drasgow (1998). The performance rating process starts by presenting the rater with two behavioral statements scaled according to effectiveness, one reflecting somewhat below-average performance and the other

somewhat above-average performance on the same dimension. The rater is asked which of the two statements is more descriptive of the ratee. This choice of one of the behavioral statements establishes the initial estimated performance level on this dimension. This choice also triggers the algorithm, which selects from the statement pool two different statements that straddle the initial estimated performance level but do so such that the next pair of statements will result in maximizing (in an item response theory sense) the amount of information to be derived from the rater’s next selection of a statement. This process is iterative and can proceed for several additional steps.

In a laboratory study, where raters evaluated videotaped ratees’ scripted performance on each of the three dimensions using CARS and either a graphic rating form or a behaviorally anchored format (BARS), the CARS had 23–37% lower standard error of measurement and higher validity against target scores represented in the videotaped performances with an overall effect size of .18. The purpose here is to explore possible reasons for these findings with CARS, and by extension, why the ideal point model might provide more precise and valid measurement in other applications such as personality testing.

In our research, we considered several possible reasons for the favorable results

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with CARS. These included: (a) CARS may simply include more behavioral information than BARS or other rating formats, and (b) CARS focuses on behavior, with no numbers evident to the rater. Borman et al. (2001) found these two explanations implausible because (a) the behavior summary scale, which has anchors that summarize a relatively large number of individual critical incidents, was not found to be any better psychometrically than a BARS format (Borman, 1979) and (b) the mixed standard scale, which does not involve any numbers, did not result in superior psychometric characteristics compared to a format with numbers (Dickinson & Zellinger, 1980).

Instead, two other features of CARS have a higher likelihood of explaining the positive results for this format. First, CARS presents behavioral statements targeted directly toward the ratee's estimated performance level, whereas BARS, for example, presents the same behavioral anchors regardless of the ratee being evaluated. CARS presents pairs of behavioral statements selected on the basis of a continuously updated estimated performance level for the ratee on each dimension. Thus, the adaptive feature of CARS is a more likely reason for the favorable results of CARS.

A second feature of CARS that is likely a contributing factor to the favorable results is the paired-comparison format itself. The cognitive processes related to comparing observed behavior with two behavioral statements and the forced choice that is required may be a comparatively effective way to generate evaluative judgments. What might be useful in considering this possible explanation is to revisit Thurstone's (1927) law of comparative judgment, one of the cornerstones of Zinnes and Griggs's (1974) model, on which the CARS is based. Thus, the CARS can be viewed as converting raters' judgments of ratee effectiveness to an interval scale on the basis of the same principles. Essentially, the choice of one behavioral statement over the other provides information concerning the placement of each target person (i.e., the ratee) on the

underlying dimension at the interval scale level.

Because it does not use paired-comparison judgments, BARS is unlikely to provide this type of interval scaling information. The scaling provided by CARS can be seen as more accurately reflecting the underlying measurement scale that raters are using to make their judgments. This scaling, in turn, may be what leads to more reliable, and valid, ratings. Of course, the BARS and other formats do not prevent raters from providing interval-level information, but this information will be at the interval level only if raters are able to provide ratings that directly reflect the true underlying scale of the attribute being rated. CARS does not assume that raters are able to make ratings on a true interval scale. Rather, CARS uses a scaling model to transform rater responses and place them on the appropriate interval scale.

On balance, the most likely explanations for the favorable results regarding CARS are (a) the CARS format presents behavioral statements more and more targeted directly toward the ratee (i.e., the adaptive feature of CARS) and (b) the iterative paired-comparison judgment task places ratee performance on an interval scale more so than the judgment tasks associated with other formats such as BARS.

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The Ideal Point Model in Action: How the Use of Computer Adaptive Personality Scales Benefits Organizations

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In their final thoughts, Drasgow, Chernyshenko, and Stark (2010) state that although the ideal point model holds much promise for improved assessment instruments, there are relatively few applications of the approach. We and more than 50 scientists from Personnel Decisions Research Institutes, PreVisor, and Navy Personnel Research, Studies, and Technology have applied the ideal point assumptions of the item response process to the development of a computer adaptive testing (CAT) approach to the measurement of personality traits. On the basis of the algorithm underlying the computer adaptive rating scales method (CARS; Borman et al., 2001), this approach leverages advances in measurement and technology to efficiently assess employees' traits and characteristics. These efforts have led to the development of new personality instruments for use in the military and private sectors. In this commentary, we discuss the practical benefits of CAT-based personality measurement and identify areas for further research inquiry.

This line of research started with the military's desire to augment the sole

selection tool, the Armed Services Vocational Aptitude Battery. Recognition that individuals' personal characteristics are useful for identifying who will be best suited for future military missions led to the development of the Navy Computer Adaptive Personality Scales (NCAPS; Borman, Houston, Schneider, & Ferstl, 2005; Houston, Borman, Farmer, & Bearden, 2005; Schneider et al., 2006). The successful validation and implementation of computer adaptive personality scales for the Navy led to the subsequent development of a set of scales for the private sector, GPI-Adaptive.

CAT Personality: How It Works

In brief, computer adaptive testing principles are used to estimate examinees' personality trait levels through an iterative, item response theory (IRT) driven, paired-comparison assessment process. The IRT model is based on the ideal point paired-comparison approach defined by Zinnes and Griggs (1974) and developed further by Stark and Drasgow (1998). Examinees select which of the two statements representing different levels of a single personality trait is more descriptive of them, and are then presented with two additional statements, using an updated trait level estimate based on their previous responses. Sequences of statement pairs are selected in a manner that maximizes information in an IRT sense,

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as only statements that are relevant to understanding an examinee's level on a given trait are presented. Presentation of statement pairs for a given trait terminates when either a sufficiently low standard error of measurement is reached or a predetermined maximum number of statement pairs has been presented.

Practical Advantages of CAT-Based Personality Measurement

CAT-based personality measurement offers many potential benefits, including increased efficiency, potential for improved validity, and increased test security. It also holds promise for addressing some of the long-standing challenges associated with personality measurement, most notably the opportunity for faking (Underhill, Lords, & Bearden, 2006), a topic that has received much attention and debate in recent years (Morgeson et al., 2007). Our experience has indeed shown this to be the case. To date, we have implemented CAT-based personality measures with more than 30 organizations and have been able to demonstrate clear benefits compared to static assessments that use Likert-type response scales.

Appropriateness for Unproctored Testing

With unproctored Internet testing (UIT), one of the greatest challenges is that of the security of the test content in order to provide fair testing conditions to all candidates. The risks associated with UIT are greatly mitigated through the use of CAT, as large item pools make it virtually impossible for candidates to collude and gain access to all (or even a significant portion) of the test items. In our approach to adaptive personality testing, CAT creates a unique testing experience for each candidate, thereby greatly increasing the security of the assessment and reducing the opportunities for cheating as a result of pre knowledge of the test questions. Specifically, statements that are targeted to an examinee's latent trait level are pulled from large banks composed

of more than 2,500 statements. As UIT continues to rise in popularity because of reduced resources for on-site proctoring, dynamic presentation of noncognitive content through an approach such as CAT is critical for supporting UIT goals and preserving the psychometric properties of assessment content.

Potential for Enhanced Validity

Increased measurement precision associated with CAT-based personality measurement brings the potential for enhanced validity because of better measurement across the trait level continuum. Validity evidence for CAT-based personality tools has been gathered through large-scale validation and consortium studies. Criterion-related validity research on NCAPS was conducted to investigate the relationship between a traditionally formatted version (using a Likert-type response scale) and the adaptive version and, importantly, to compare the validities of the two versions. Results indicated that the validity of the adaptive version generally exceeded that of the static instrument. In addition, intercorrelations between adaptive NCAPS scales were lower than those between traditional NCAPS, suggesting greater construct validity for the adaptive version and the potential for higher validities for composites of the scales.

Results from consortium validation efforts conducted on GPI-Adaptive indicate that the validity is on par or exceeds that of traditional personality instruments reported in the literature. Furthermore, given the modular nature of the assessment (such that the most job relevant and valid scales can be assembled to create tailored test batteries), we have found uncorrected composite validity coefficients to exceed the "personality coefficient" of .30. This modular approach to testing facilitates a "whole person" approach by efficiently combining tests that measure an array of competencies.

Reduction in Testing Time

Reduced testing time is another key benefit of CAT-based personality tools, which

is critical particularly as organizations seek to streamline selection processes and minimize candidate dropout rates. Most comprehensive personality assessments typically involve administration of hundreds of items to reliably measure personality traits. CAT-based personality measurement allows for precise measurement with fewer items compared to traditional personality instruments. The assessment terminates when sufficient information about a test taker has been gathered because only items that contribute information about a candidate's trait level are administered. Research conducted on GPI-A indicates that the testing time for the adaptive version requires one third less time than an equivalent static measure.

Reduction in Vulnerability to Faking and Impression Management

Concerns about faking on personality tests have peaked in recent years (Morgeson et al., 2007), leading researchers to call for greater innovation in the item response format. The forced-choice item format has been suggested as a promising item type to help reduce opportunity for faking. The forced-choice format, even our version that draws statements from the same dimension, may remove many of the cues that a respondent could use to accomplish faking. With a standard Likert-type response format, only the direction of a personality item needs to be determined in order to make an extreme response in that direction. With the paired-statement format, there is no cue from the response scale itself. Preliminary research (Underhill et al., 2006) indicates that these types of items are more difficult to distort. Specifically, Underhill et al. (2006) found that, when prompted, test takers were not able to significantly elevate their scores to as large an extent on the adaptive NCAPS when compared to traditional NCAPS.

Research Agenda for CAT-Based Personality Measurement

As organizations report a greater need for efficiency and convenience in their

selection processes, it is incumbent upon industrial–organizational (I–O) practitioners and researchers to identify methods that meet these needs without sacrificing validity and test security. We suggest that CAT-based personality measurement is one way I–O psychologists can meet these challenges, but work remains to refine the approach. We propose a set of research questions that will inform future test development, implementation, and adoption of these tools.

- *Faking susceptibility.* We need more research on CAT-based personality tools and the extent to which candidates can and do fake on these types of measures. Findings to date need to be replicated with additional measures and comparing item pairs that are unidimensional to those that are multidimensional.
- *Optimizing prediction by exploring nonlinear relationships to criteria.* It is compelling to explore whether the greater precision in CAT-based measurement would enhance our ability to discover nonlinear relationships between various personality constructs and job performance. This could result in substantial gains in operational validity if a nonlinear scoring approach to these measures was used.
- *Efficiency and effectiveness of CAT-based personality measurement.* Finally, compared to static instruments, to what extent does adaptive personality testing result in an improved candidate experience and better outcomes for recruiters and hiring managers? More studies are needed to support preliminary findings on testing times, dropout rates, and candidate reactions to CAT-based personality tests.

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Toward Understanding the Psychology of Unfolding

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The ideal point or unfolding conception of psychological responses to personality and attitudinal items offers many advantages over the dominance model, as outlined by Drasgow, Stark, and Chernyshenko (2010). Past research involving those and other authors (Chernyshenko, Stark, Drasgow, & Roberts, 2007; Stark, Chernyshenko, Drasgow, & Williams, 2006; Weekers & Meijer, 2008) has demonstrated empirically the utility of unfolding models. However, the psychology of unfolding models has yet to be addressed directly in the published literature. In this commentary, we present an agenda toward an increased understanding of the psychology of unfolding items by describing some of our own research in progress as part of the Personality Measurement Research Group at Bowling Green State University.

Drasgow et al. propose that introspection is the mechanism that causes responses to fit the unfolding model, whereas items that do not require introspection should be fit better by the dominance model. We agree that introspection is a key component for differentiating between the types of items that would require one model versus the other. However, we believe that the

mechanisms behind unfolding items are much more complicated than just introspection or not. We address three issues involved in unfolding models: (a) individual differences in the use of the ideal point process, (b) item/stimulus content and administration modality, and (c) the nature of the attribute being measured.

Individual Differences

Research has demonstrated that there are individual differences in the processes that people use to respond to psychological items, and we expect that individuals will differ on whether they are likely to use the dominance or ideal point processes. We also believe based on some prior experiences that many individuals have been trained to respond to items in a dominance perspective and will do so regardless of the content of items. In one survey administration, we had a respondent raise his hand and demand to know how he was supposed to respond to the item “I like to go to parties some of the times.” He was frustrated because the quantifier “some of the times” was odd from his perspective, and he was unsure how to fit his level of extraversion onto this item. Meanwhile, the rest of the students completing the survey had no issues and responded to all items in a quick fashion.

We developed a measure of literal-mindedness to capture the individual difference that likely caused that respondent to become frustrated. Individuals who are high in literal-mindedness will respond to the

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content of items, whereas people who are low in literal-mindedness will respond in ways that they think the test makers intend. In one study (Lin et al., 2009), we found that those who responded to the overt literal wording in the items on a workplace attitudinal scale (Workplace Drug Testing) were less likely to follow the expected response pattern for unfolding than were those who attempted to interpret the deeper meaning of items.

In a follow-up study (Lake & Zickar, 2010), we administered two measures of mental effort: one assessing the amount of thought put into responding and the other assessing the amount of cognitive effort expended. Small relationships emerged between thought, effort, and person misfit similar to that of Lin et al. (2009). Respondents with extremely high and extremely low thought and effort scores were less likely to follow the expected response pattern for unfolding than those with moderate scores. Taken together, the results of these two individual differences studies suggest that a moderate amount of item interpretation and thought leads individuals to follow the expected response pattern for unfolding, whereas too much or too little interpretation may lead to divergence from this response pattern.

Item Content and Administration

In addition to individual differences, we believe that more research is needed to understand the features of items that relate to unfolding. The published research provides anecdotal evidence about the features of items that lead to unfolding, although more research is needed to understand features that cause some items to unfold and others not to. On the basis of studies of context effects in scales, we believed that respondents might look to nearby scale items when they encountered confusing intermediate items. Looking at nearby extremely worded items might promote the use of a dominance (rather than unfolding) response pattern.

In an attempt to reduce the potentially biasing effects of nearby items, we

administered the attitudinal scale items to participants one at a time online (Lake & Zickar, 2010). Respondents in this experimental condition were only able to view the item they were currently responding to, thereby reducing any biased responding occurring as a result of a respondent looking at nearby items. Results from this administration were compared to results obtained from a paper-and-pencil administration where respondents were presented all the items in block format and could easily look at nearby items. At the item level, the results of this manipulation were mixed; some items showed increased unfolding in the experimental condition, whereas others showed decreased unfolding. At the scale level, however, there appeared to be some advantage to administering items one at a time. Model-data fit appeared to be better in the experimental condition. Additionally, the scale had a smaller standard error of measure in the experimental condition across most of the attitude continuum.

Some items used by Chernyshenko et al. (2007) that were unfolding contained words we refer to as "vague quantifiers" (e.g., "not exactly," "about," "not always;" see p. 100, Table 4). Past research has shown vague quantifiers to have very different meanings to different people, leaving room for interpretation (a similar rationale to the development of our literal-mindedness construct). Therefore, we wondered whether the use of vague quantifiers in typical personality items would be helpful in forcing items to unfold by making them appear less extreme in wording.

To understand the influence of vague quantifiers on item location, we analyzed data from three groups, the first a control group taking 20 Conscientiousness items from the International Personality Item Pool (IPIP; e.g., "Am always prepared"). For each item, we determined a way to alter it with the word "sometimes," attempting to change little or nothing else about the item (e.g., "Am sometimes prepared"). One group took a form where the manipulation (i.e., those with "sometimes" in the stem) was used on the even-numbered items;

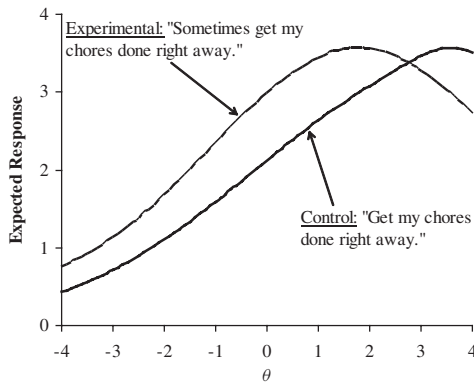


Figure 1. Item 4, International Personality Item Pool Conscientiousness for the control and experimental groups.

another took a form with “sometimes” in the odd-numbered items (Carter, Dalal, & Zickar, 2010).

Figure 1 shows the item “Get my chores done right away” when not manipulated and an overlay of the item when it reads “Sometimes get my chores done right away.” Comparing these curves, we can see the manipulation worked. In fact, we found that the manipulation showed shifts toward unfolding with medium to large effect sizes. However, we found that only about half the manipulated items in either group showed unfolding to a level that was significant. This suggests that although vague quantifiers tend to make items less extreme, there is more to writing an unfolding item than just sticking the word “sometimes” in the stem (Carter et al., 2010).

Taken together, we believe these results indicate that item content and administration modalities show some covariation with the extent to which persons’ responses follow an unfolding model. Of course, these studies have limitations in generalizability; however, they are explorations that may hint at future possibilities for uncovering how items unfold.

Esoteric Constructs and Rating Targets

In addition to the people and the content involved in measuring attributes, the nature

of the construct itself could have an influence on the unfolding of items. In other words, what is the influence of the nature of the thing we are asking respondents to rate? For example, attributes differ from one another in regard to how esoteric or abstract they are. We believe that measures of more esoteric and abstract attitudinal and personality constructs may show more unfolding than more grounded, realistic constructs.

One particular study spurned our thinking in this regard. Carter and Dalal (2010) analyzed the five scales of the Job Descriptive Index (JDI), a measure of job satisfaction, in which persons respond Yes or No to adjectives such as “Boring.” Of the five facet scales, only the Work Satisfaction scale showed a considerable number of unfolding items. Comparing this scale to the others in the JDI (pay, promotions, supervisor, and coworkers), we thought it striking that of these facets work seems to be different than the others in that ratings on the items for the work scale have no clear target. In other words, the concept of work may be more abstract or esoteric than the others.

One of the current authors (Carter) is currently working on a project that involves scaling a series of traits on the continua of esotericism and abstractness. For example, Conscientiousness may be viewed as more grounded and realistic than other constructs such as Machiavellianism. Additionally, it may be that certain items are more removed from reality. This study will investigate whether more esoteric or abstract traits and items tend to have a larger proportion of unfolding items in current IPIP measures by examining whether respondents’ post hoc judgments of items and traits are correlated with the number of items that show unfolding in a scale. We also plan to investigate whether persons who view certain traits or items as esoteric or abstract show response patterns that are not what would be expected in unfolding items.

Conclusions

These three streams of research into the determinants of unfolding are by no means

exhaustive or comprehensive. As Drasgow et al. have noted, the reemergence of ideal point models via item response theory modeling has opened an exciting and interesting area of inquiry. Much more research is needed to understand when unfolding can be expected and more appropriate than dominance models.

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RESPONSE

Improving the Measurement of Psychological Variables: Ideal Point Models Rock!

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Abstract

Although there is no doubt that Likert scaling suffices for straightforward scale development and use, it is important to appropriately model the response process for more complex measurement problems. In this response, we comment on the response process and four applications: assessment of dimensionality, computerized adaptive testing, differential item functioning, and individual differences in responding. In each case, we argue that correctly modeling the psychology of responding is critical.

We would like to thank the commentators for providing many interesting and useful ideas. They note some important research contributions that have already been made and describe a large number of significant questions and topics for future research. Taken together, the commentaries provide a strong impetus for improved measurement and better research.

At the outset of our response to the commentaries, we want to reiterate our statement that Likert scaling works well as

a rough and ready measurement method for a large proportion of noncognitive items. Likert scales are relatively easy to construct, reliable, and have been shown to be valid measures of many important psychological constructs. For rough and ready applications, little may be gained from the use of the more complex and demanding ideal point models. Thus, for straightforward applications, we do not disagree with Waples, Weyhrauch, Connell, and Culbertson (2010) that Likert scaling suffices.

But what about more complex applications? In our response, we comment on five: identifying underlying response processes, studying dimensionality, computerized adaptive testing (CAT), differential item functioning (DIF), and individual differences in responding.

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The Underlying Response Process

Reise (2010) agrees that it is critically important to examine and understand the process that underlies item responses. In addition to being of fundamental psychological importance, psychometric models and analyses should be based on appropriate assumptions of the response process. As noted in our discussion of dimensionality, the application of an inappropriate psychometric analysis can produce misleading results.

Likert scaling is not explicitly derived from a psychometric model; in contrast, Thurstone's 1928 article shows how a measurement procedure can be based on assumptions about a response process. Importantly, the steps in Likert scaling only make sense if a dominance process underlies responses. For example, item-total correlations, which Likert (1932) stressed as a key component of his approach, are informative about the quality of items for a dominance response process but not an ideal point process.

What is the process underlying responses to items assessing attitudes, values, personality, vocational interest, and other important psychological constructs? In our focal article, we provided a conceptual argument for an ideal point process. Specifically, when introspection is required, we argued that people consider the extent to which an item describes them. If the item provides a reasonably accurate description, then they agree; otherwise, they disagree.

How can we empirically test this hypothesis? In our work, we have used large data sets and fit both ideal point and dominance models. We have then examined the goodness of fit by a chi-square statistic that compares observed frequencies to those predicted by the fitted models. The approach of Brown and Maydeu-Olivares (2010) is similar; they inspect the items and make judgments as to whether each is likely to be answered by an ideal point or dominance process. Then they fit unidimensional and multidimensional dominance and ideal point models.

Drasgow, Levine, Tsien, Williams, and Mead (1995) provide details about the chi-square statistics that we use. They are based on the assumption that a model provides a good description of the data when it accurately reproduces the observed frequencies. For n dichotomously scored items, there are 2^n cells in the n -way contingency table of observed responses. Unfortunately, in practice a table with 2^n cells is too large: Many cells will be empty. Consequently, we examine lower order subtables to assess fit.

Reise (2010) expresses concern about this approach to assessing fit; we also worry about its efficacy. In a series of simulation studies (Nye, Guo, & Drasgow, 2010; Tay, Ali, Drasgow, & Williams, 2010), we have studied the performance of several fit statistics. We have found that some statistics have very little power to detect misfit (e.g., infit, outfit, BILOG's chi-square, Yen's Q3, and Drasgow et al.'s [1995] chi-square for single items). Our preferred approach to assessing fit, Drasgow et al.'s chi-square for pairs and triples of items, was found to be effective, as was an index proposed by Orlando and Thissen (2000). Moreover, we have found that it is possible to improve on the Drasgow et al. method, particularly when sample sizes are not very large. Joe and Maydeu-Olivares (2006) have derived the sampling distribution of the Drasgow et al. chi-square fit measure. Although their equations are complex, their work suggests that further improvements and refinements are possible.

Unfortunately, these fit statistics are influenced by both violations of local independence (e.g., pairs of items with overly similar content and other types of multidimensionality) and misspecified item response functions. Thus, a finding of lack of fit does not unequivocally prove that the hypothesized response process is misspecified. Consequently, additional analyses should be conducted to examine dimensionality. In such analyses, we have found many personality scales to be relatively unidimensional. This has led us to believe that the culprit for the lack of fit

of dominance models lies in a misspecified response process. In any case, we believe new ideas and new approaches to studying whether a psychometric model adequately fits data are greatly needed.

Dimensionality

Davison (1977) showed that factor analysis of unidimensional ideal point data can produce a second artifactual dimension. Spector and Brannick (2010) (see also Spector, Van Katwyk, Brannick, & Chen, 1997) have extended and elaborated on this very important point. These papers show why it is so important to understand the underlying response process: If it is misspecified, subsequent statistical analyses may provide misleading or erroneous results.

Although Spector et al. (1997) clearly articulate problems that can arise from applying dominance model analyses to ideal point data, Credé (2010) warns about problems resulting from the reverse situation: applying ideal point models to dominance data. He notes that an attitude can be bipolar, ranging from negative to positive, or bivariate, with two unipolar axes. In the bivariate case, one axis varies from neutral to highly positive and the other axis varies from neutral to highly negative. In this bivariate framework, an individual can be ambivalent in that he or she is simultaneously positive and negative in attitude.

How can we test whether an attitude is bivariate or bipolar? As noted previously, Brown and Maydeu-Olivares (2010) fit unidimensional and multidimensional dominance and ideal point models. If the attitude is bivariate, we would expect the multidimensional models to provide substantially improved fit. Brown and Maydeu-Olivares note an unfortunate complication to an already thorny problem: It may be inherently more difficult to fit ideal point models than dominance models. For example, they compared the normal ogive dominance model to an ideal point counterpart and found much more accurate item parameter estimates when the dominance model was fit to dominance data than when the ideal

point model was fit to ideal point data. Again, it is clear that better psychometric tools are needed.

Computerized Adaptive Testing

In CAT, an item is drawn from an item pool for each person so that its difficulty (in the case of ability testing) or extremity (in the case of personality assessment) is appropriate. After the item is administered, the person's standing on the latent trait is estimated, which leads to selection of the next item. This iterative process ordinarily continues until a highly precise latent trait estimate is obtained.

We have not conducted simulation studies to examine the problems that might arise from using a dominance model for CAT when the true response process is ideal point. But two of the commentaries describe highly successful implementations of CATs based on ideal point measurement.

In the first, Borman (2010) describes one of the most striking improvements in the measurement of job performance in the half century since the introduction of behaviorally anchored rating scales. In their summary of research on rating scales for performance measurement, Landy and Farr (1980) found so little progress that they called for a moratorium on research in this area. Borman's computerized adaptive rating scale (CARS) circumvents the problems of rating scales by presenting two statements to the rater who simply picks the one that is more descriptive. Borman reports 23–37% lower standard errors of measurement and higher validity for assessments of performance obtained with CARS.

When a rater is given two statements and considers which is a more accurate description of an employee, it seems clear that an ideal point response process is utilized. The CARS model assumes that the rater perceives some level of performance for the ratee and then mentally evaluates the difference in the level of performance corresponding to each statement to the ratee's level. It seems reasonable to assume that the levels of performance associated

with the ratee and the two statements are perceived with error, which adds to the complexity of the psychometric model needed for CARS.

Kantrowitz and Tuzinski (2010) describe two personality CATs, the Navy Computer Adaptive Personality Scales and the GPI-Adaptive, that use the same underlying ideal point model (the Zinnes and Griggs' [1974] model) as CARS. They report uncorrected validities of composites that exceed .30, which has long been viewed as an upper limit. Moreover, by developing a very large item pool (over 2,500 personality statements), it is virtually certain that each person will receive a unique set of items. This is quite important in the context of unproctored Internet assessment.

Differential Item Functioning

What would happen if a process underlying item responses was ideal point but a researcher conducted DIF analyses based on dominance models? We do not know of research investigating this situation. However, if the model used to describe responses in each of the two groups is misspecified, comparisons across the two groups seem inherently problematic. We suspect that it would be very difficult to control the Type I error rate while maintaining power to detect true differences. Research in this area is greatly needed.

Individual Differences in Responding

Even though a test or assessment may have high reliability, some individuals may be poorly measured (Levine & Drasgow, 1982). There has been quite a bit of research on this topic, called appropriateness measurement or person fit, during the past 3 decades. Two of the commentaries, both directed at understanding the psychology of unfolding, present ideas that can be related to appropriateness measurement.

Oswald and Schell (2010) argue that it is important to understand the psychology of responding to items with intermediate

extremity. They provide three explanations for why a person might endorse such an item: The person has an intermediate standing on the trait being assessed, the person has a highly nuanced interpretation of the trait being assessed so that his or her judgment is highly contingent on specific features of the items, and the person is indifferent to the trait being assessed and consequently his or her responses are essentially random. The first interpretation is likely to be consistent with an ideal point model fit to a group's data, but the second two describe atypical responding. A person fit index might be able to identify individuals responding according to interpretations two and three. But little or no work has been done on person fit in the context of ideal point responding, so new research is needed.

Carter, Lake, and Zickar (2010) do not disagree with ideal point modeling but argue that our use of introspection as the explanation is too simplistic. For example, they expect that people may differ in whether they respond to a given item by an ideal point or dominance process. They have found that individuals exerting extremely high or extremely low effort in responding were apparently less likely to answer via an ideal point process.

One of the most interesting new trends in psychometrics—mixture models—may provide tools to test Carter et al.'s (2010) ideas. These models assume that populations are not homogeneous but instead are composed of a small number of subpopulations that respond according to their own submodels. Rost's (1991) mixture model is an early example and, more recently, Vermunt and Magidson's (2005) Latent GOLD software has greatly increased the range of mixture models that can be fit.

Final Thoughts

Dalal, Withrow, Gibby, and Zickar (2010) present six questions that practitioners and others might have about ideal point items and models. Their questions are interesting and important. Unfortunately, Dalal et al.

(2010) did not provide answers to the questions. One question that Dalal et al. (2010) did not ask is "Are ideal point models a panacea for all your measurement problems?"

We can provide an answer to this question, and it is "no." As Brown and Maydeu-Olivares (2010) note, it is difficult to write good items with intermediate extremity parameters. Then fairly large samples (perhaps 500 or more) are needed to estimate the item parameters of ideal point models. We have also found that sometimes the estimation software is recalcitrant and numerous runs are needed before reasonable results are obtained. For example, with some computer programs, it is necessary for a user to provide a rough ordering of items from low to high trait levels to obtain starting values for parameter estimation; others use diagnostic statistical tests to accomplish this objective. In both cases, it is frequently necessary with models involving multiple parameters per item to impose equality constraints to attain convergence on initial runs, thus fitting more restrictive models until a viable subset of items can be identified for calibration with a more general model. Because the item parameter estimates for some models are also substantially correlated, it can be difficult to predict how item response and item information functions will look without plotting, so constructing scales that provide high information at targeted trait levels can be a cumbersome, data-driven process. Fortunately, automated test assembly and adaptive item selection algorithms offer potential solutions to this problem for large-scale testing applications, and item response theory (IRT) based trait scores appear to be robust to modest degrees of error in item parameter estimates (Stark, Chernyshenko, & Guenole, in press).

In conclusion, we believe that despite the technical difficulties currently associated with the application of ideal point methods, the shift away from exclusive reliance on dominance models is a step in the right direction. It is simply not possible to fit statements representing moderate

("neutral") trait levels with traditional models. The item-total correlations and logistic IRT discrimination parameters are often near zero for statements such as "My social skills are about average" and "My life has had an equal share of ups and downs," even though we believe that such statements capture the essence of their respective behavioral domains. Moreover, with research having shown that ideal point models can be fitted successfully to scales containing a preponderance of positive and negative statements, which exhibit largely s-shaped item response functions and satisfy traditional scale construction criteria (Chernyshenko, Stark, Drasgow, & Roberts, 2007; Stark, Chernyshenko, Drasgow, & Williams, 2006), we ask rhetorically why one would not continue exploring models that allow us to do more with our item pools rather than less. If ideal point models can gainfully utilize a higher proportion of statements written by subject-matter experts, which often requires substantial time, effort, and cost, and thus provide better representation of behavior across the range of a trait continuum, then the return on investment in item development and measurement precision should outweigh the technical challenges encountered with the use of currently available software.

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